Optimal Light Spectra Under Regulatory Constraints: How Efficiency Standards Drive Lamps to Weird Spectral States

Generated by `photopic` — a Modula-3/Scheme spectral optimizer

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Abstract

Government regulations on lamp efficiency (lumens per watt) and color quality (Color Rendering Index) define a constrained optimization problem whose solutions are not the smooth, broad spectra of incandescent bulbs but narrow-band, spiky spectral power distributions. We use `photopic`, a hybrid Modula-3/Scheme numerical tool, to explore the Pareto frontier of this trade-off. As the efficacy requirement increases, the optimal spectrum converges to approximately three narrow peaks near 450, 540, and 610 nm — Thornton's "prime color" wavelengths — while aggressively suppressing power at all other wavelengths, including the red region critical for color fidelity. The results show that regulatory pressure, when pushed to its logical extreme, drives light sources toward spectra that are visually "weird": they satisfy the letter of color rendering standards while producing light that many people find unnatural.

1 Introduction

An incandescent lamp is a near-perfect blackbody radiator. Its spectrum is broad and smooth, rendering colors with perfect fidelity (CRI = 100). But most of its radiation falls in the infrared, giving it a luminous efficacy of only ∼12 lm/W. Paradoxically, incandescent lamps are excellent *radiant* emitters: a tungsten filament at 2700 K converts roughly 85% of its electrical input into radiation. The problem is that at this temperature only ∼5% of the blackbody spectrum falls in the visible band; the rest is infrared. Government regulations worldwide — EU Ecodesign, US EISA/DOE, California Title 24 — have mandated minimum efficacy standards that incandescent lamps cannot meet, effectively banning them.

The replacement technologies (fluorescent, LED) achieve high efficacy by concentrating their emission in the visible spectrum. But *where* in the visible spectrum? The CIE photopic luminous efficiency function $V(\lambda)$ peaks sharply at 555 nm (yellow-green). A monochromatic source at 555 nm with perfect electrical-to-optical conversion would achieve the theoretical maximum of 683 lm/W — but it would render no colors at all.

Legislators recognized that pushing efficacy alone would produce light that people hate, so they added color quality requirements. The instrument they chose is the CIE Color Rendering Index (CRI), a 1965-era metric based on only 8 test color samples. The regulatory question thus becomes: **What spectral power distribution maximizes luminous efficacy while satisfying the CRI constraint?**

This is a constrained optimization problem. The constraints come from regulations:

- **CRI $\geq$ 80** (Energy Star, EU); CRI $\geq$ 90 (California Title 24 for some applications)
- **CCT within ±50K** of the nominal color temperature (e.g., 2700K for "warm white")

- **Duv $\leq$ 0.012**: chromaticity tolerance (distance from the blackbody locus)

The central question of this report is whether the legislative response actually solves the problem that people care about — faithful color reproduction — or whether CRI's 8-sample design leaves enough room for an optimizer to "game the metric," producing spectra that pass the regulatory test while rendering many real-world colors poorly. We explore what happens as you turn up the efficacy requirement and let the optimizer find the best spectrum that satisfies CRI. The answer is that the optimal spectra become increasingly bizarre, and CRI fails to prevent this.

2 Background

2.1 Luminous Efficacy

Luminous efficacy of radiation (LER) is the ratio of luminous flux to radiant power:

$$\eta = K_m \frac{\int_0^\infty S(\lambda)\, V(\lambda)\, d\lambda}{\int_0^\infty S(\lambda)\, d\lambda} \tag{1}$$

where $S(\lambda)$ is the spectral power distribution, $V(\lambda)$ is the CIE 1931 photopic luminous efficiency function, and $K_m = 683$ lm/W is the maximum luminous efficacy at 555 nm.

For a 2700K blackbody, the total LER is only $\sim$15 lm/W because most radiation is in the infrared. If we restrict to the visible band (380–770 nm), the visible-only LER is $\sim$88 lm/W.

2.2 Color Rendering Index (CRI)

The CRI quantifies how faithfully a light source renders colors compared to a reference illuminant (a blackbody at the same correlated color temperature). The calculation, following CIE 13.3-1995 and the colorimetric formulas in Foley and van Dam [5], involves:

1. Normalize the test spectrum to $Y = 100$
2. Find the reference blackbody at the correlated color temperature (CCT) via search in CIE 1960 MacAdam (u, v) space
3. For each of 14 Test Color Samples (TCS), compute the color shift ΔE^*_{UVW} using von Kries chromatic adaptation
4. $R_i = 100 - 4.6\, \Delta E^*_{UVW}$ for each sample; $\mathrm{CRI}(R_a) = \frac{1}{8} \sum_{i=1}^{8} R_i$

The key subtlety: only TCS 1–8 contribute to R_a. TCS 9 (saturated red) is *not* part of the official CRI score. This creates a loophole that the optimizer exploits.

2.3 Thornton's Prime Colors

In 1971, W. A. Thornton of Westinghouse showed that the most chromatically effective wavelengths for white light are three narrow bands near **450, 540, and 610 nm** — the "prime colors" [1]. Conversely, wavelengths near 500 nm and 580 nm are "anti-prime" — they contribute power without adding chromaticity discrimination. Thornton was named US National "Inventor of the Year" in 1979 for his prime-color fluorescent lamp patents [4].

The optimization results below independently rediscover Thornton's prime colors from first principles.

3 What Gets Regulated as a "White Lamp"?

Not every light source falls under efficiency and color-quality regulations. Colored lamps, decorative filament bulbs, stage lighting, plant grow lights, and various specialty products are typically exempt. The regulations target *general-service lamps* — the everyday bulbs that replaced incandescents in homes and offices. The precise definition of "in scope" varies by jurisdiction and determines which products face the efficacy/CRI constraints that drive the optimization problem analyzed in this report.

3.1 EU Ecodesign (2019/2020)

The EU regulation defines its scope via a *chromaticity boundary* rather than an explicit CCT range. A light source is treated as white (and thus in scope) if its chromaticity coordinates satisfy:

$$0.270 < x < 0.530 \quad \text{and} \quad -2.3172\,x^2 + 2.3653\,x - 0.2199 < y < -2.3172\,x^2 + 2.3653\,x - 0.1595$$

On the Planckian locus, $x = 0.530$ corresponds to approximately 1800 K, so the regulation covers very warm lamps. Sources outside this boundary (UV, deep red, amber, grow lights) are exempt. Quality requirements: CRI ≥ 80, with an exemption for HID >4 klm and outdoor/industrial applications. There is no R_9 requirement and no TM-30 requirement.

3.2 US DOE General Service Lamp (GSL)

The US defines scope differently: a lamp with CCT < 2500 K is classified as "colored" and falls outside the general-service lamp definition under EPCA, exempting it from efficiency standards. Lamps with CCT ≥ 2500 K and lumen output between 310 and 3300 lm are in scope. The current backstop (2022) requires only 45 lm/W; the 2028 standard raises this to ∼125 lm/W for a typical 810-lumen omnidirectional lamp. The DOE imposes *no color quality requirements at all* — neither CRI nor R_9.

3.3 California Title 24 JA8

JA8 is the most stringent regime: CCT $\leq$ 4000 K (warm and neutral white only), CRI ≥ 90, $R_9 \geq 50$, minimum efficacy 45 lm/W. Products with CCT $>$ 4000 K are not JA8-eligible for high-efficacy residential applications.

3.4 ANSI C78.377 Chromaticity Standard

While not itself a regulation, ANSI C78.377 defines the nominal CCT categories that manufacturers target and that regulations reference. The 2024 revision added 1800 K and 2000 K for the first time, reflecting the growing market for "ultra-warm" and "candlelight" LED products. The standard now covers 10 nominal CCTs: 1800, 2000, 2200, 2500, 2700, 3000, 3500, 4000, 4500, 5000, 5700, and 6500 K.

3.5 Summary and Implications for This Study

Table 1: Regulatory scope: which lamps must meet efficiency and color standards.

Jurisdiction	Min CCT	Max CCT	CRI	Notes
EU 2019/2020	∼1800	∼10000	≥80	Chromaticity boundary
US DOE (2022)	2500	—	—	45 lm/W backstop
US DOE (2028)	2500	—	—	∼125 lm/W, no CRI
CA Title 24 JA8	—	4000	≥90	$R_9 \geq 50$
ANSI C78.377	1800	6500	—	Chromaticity bins only

3.6 The Regulatory War on Coziness

Consumers choose low CCT for ambiance. 2700 K is the standard residential warm white; 2200 K is marketed as "warm dim" or "vintage"; 2000 K and 1800 K appear as "candlelight" or "ultra-warm" in restaurants, bedrooms, and hospitality settings — precisely the environments where people care most about how light *feels*. The lowest CCT that falls under a binding efficiency regulation is approximately 1800 K (EU, via the chromaticity boundary) or 2500 K (US DOE).

Unfortunately, the physics of warm light works against efficiency. The warmer and cozier you want the light, the harder it is to produce efficiently:

1. **Lower theoretical LER.** The visible-band LER of a blackbody drops from ∼109 lm/W at 3000 K to ∼88 lm/W at 2700 K to ∼60 lm/W at 2200 K. Achieving a given LER floor requires more aggressive spectral shaping at lower CCT.

2. **Higher Stokes loss.** Matching the warm chromaticity requires substantial red phosphor emission (>600 nm), which incurs the largest Stokes penalty in the Philips cost model: a photon at 630 nm costs $C = 630/(450 \times 0.90) = 1.56$ — over 50% more electrical power per optical watt than the blue pump.

3. **More demanding CRI reference.** The CRI calculation compares the test spectrum against the blackbody at the measured CCT. At low CCT, this reference is very red-rich, so any deficit in deep-red emission is penalized more severely in the R_i scores.

Efficiency regulations therefore bite hardest on exactly the products that serve the most atmosphere-sensitive applications. A restaurant owner who wants 2200 K candlelight-quality LEDs faces a tighter physics budget than someone lighting an office at 4000 K — yet both must meet the same efficiency floor.

Of the major regulatory regimes, the EU is the most difficult to satisfy at low CCT, because it is the only one that simultaneously covers very warm lamps (down to ∼1800 K), requires CRI ≥ 80, *and* imposes a high efficacy floor (∼98 lm/W system). The US DOE has a higher efficacy target (125 lm/W by 2028) but imposes no CRI requirement and excludes lamps below 2500 K. California JA8 has the strictest color quality (CRI ≥ 90, $R_9 \geq 50$) but requires only 45 lm/W. Only the EU squeezes both ends — efficacy and color quality — across the widest CCT range.

Table 2 quantifies this effect by comparing the maximum achievable efficacy across four CCTs at equal CRI floors. All runs use grid mode with 129 spectral parameters.

Table 2: Maximum achievable LER (lm/W) vs CCT and CRI floor. R_9 values in parentheses show the unconstrained red-rendering penalty at each operating point.

	1800 K		2100 K		2200 K		2700 K	
CRI	LER	(R_9)	LER	(R_9)	LER	(R_9)	LER	(R_9)
≥60	412	(−34)	438	(−27)	423	(−25)	437	(−28)
≥70	400	(−15)	422	(−9)	423	(−7)	428	(−7)
≥80	385	(+11)	404	(+21)	415	(+11)	412	(+21)
≥85	370	(+25)	389	(+34)	402	(+32)	402	(+33)
≥90	360	(+35)	386	(+36)	403	(+50)	399	(+45)
≥95	366	(+72)	381	(+77)	386	(+75)	388	(+58)

At CRI ≥ 80 (the EU minimum), the efficacy penalty for dropping from 2700 K to 1800 K is about 27 lm/W — a 7% reduction in theoretical LER, which translates directly to a 7% higher electricity cost for the same lumen output. The penalty steepens at CRI ≥ 90: 39 lm/W or roughly 10%. At 1800 K, R_9 is also consistently worse at each CRI floor, confirming that warm light forces a harder compromise between efficiency and red rendering.

Several non-monotonicities appear in the table: at 1800 K, CRI ≥ 95 yields higher efficacy (366 lm/W) than CRI ≥ 90 (360 lm/W); at 2200 K, CRI ≥ 70 matches CRI ≥ 60. Since relaxing a constraint can never worsen the true optimum, these indicate that COBYLA is finding different local minima at different constraint levels — a tighter constraint can steer the optimizer away from a bad basin. See Section 14 for planned improvements to optimizer robustness.

3.6.1 Individual R_i Breakdown

The CRI general index R_a averages only the first eight test color samples (R_1–R_8), which are all unsaturated pastels. The six "special" samples (R_9–R_{14}) — saturated reds, yellows, greens, blues, plus skin and foliage — are reported separately but carry no regulatory weight. Table 3 reveals which colors the optimizer sacrifices.

Table 3: Individual R_i scores at CRI ≥ 80 and CRI ≥ 95 across four CCTs. Values below 50 are in bold.

		CRI $\geq$ 80				CRI $\geq$ 95			
	Test color	1800	2100	2200	2700	1800	2100	2200	2700
R_1	Light greyish red	87	94	83	94	98	99	99	97
R_2	Dark greyish yellow	89	85	91	88	92	91	90	91
R_3	Strong yellow green	67	68	69	70	96	96	96	91
R_4	Mod. yellowish green	82	81	81	93	99	99	99	97
R_5	Light bluish green	96	90	93	81	94	94	93	92
R_6	Light blue	81	77	83	75	94	93	93	89
R_7	Light violet	82	77	89	93	98	97	97	95
R_8	Light reddish purple	67	68	69	70	90	92	91	86
R_9	Strong red	11	21	11	21	72	77	76	58
R_{10}	Strong yellow	68	61	71	56	83	84	83	78
R_{11}	Strong green	80	78	78	84	99	96	96	97
R_{12}	Strong blue	**29**	**7**	**6**	**15**	68	60	60	**44**
R_{13}	Light skin	86	91	84	92	91	91	91	95
R_{14}	Leaf green	76	78	77	79	94	95	96	93
Worst R_i (all 14)		**11**	**7**	**6**	**15**	68	60	60	**44**

Several patterns emerge:

1. **R_{12} (strong blue) is worse than R_9 at every CCT.** At CRI ≥ 80, R_{12} is the overall worst score at 2100 K (7), 2200 K (6), and 2700 K (15). The optimizer's blue emission is a narrow spike at the 450 nm pump wavelength — efficient for chromaticity matching but inadequate for rendering the broader reflectance of saturated blue objects.

2. **Even CRI ≥ 95 cannot save R_{12}.** At 2700 K, $R_{12} = 44$ despite CRI $= 92$. This is the worst individual score in the entire table at CRI ≥ 95, and it is far worse than the corresponding $R_9 = 58$. Blue is the blind spot of CRI-constrained optimization.

3. **1800 K does *better* on R_{12} than 2700 K.** At CRI ≥ 95: $R_{12} = 68$ (1800 K) vs. 44 (2700 K). The warm CCT chromaticity requires less blue energy, giving the optimizer more freedom in placing the blue peak.

4. **R_{10} (strong yellow) is quietly poor.** At CRI ≥ 80, R_{10} ranges from 56–71 across CCTs — always below the R_a average, but not low enough to attract regulatory attention. Strong yellows fall between the optimizer's green and red peaks, receiving insufficient spectral support.

The concentration of damage in R_9, R_{10}, and R_{12} — precisely the three most saturated test colors in the red, yellow, and blue regions — confirms that the optimizer exploits the gap between CRI's pastel test samples (R_1–R_8) and the saturated colors that matter most for visual quality. A regulation requiring all 14 R_i values above a minimum threshold would close this loophole more effectively than any improvement to the averaging formula.

3.6.2 All-R_i Floor: Effect on Maximum Efficacy

To quantify the efficacy cost of closing the R_i loophole, we add an optimization mode that constrains the *worst* of all 14 individual R_i scores above a threshold, while maximizing Philips-corrected LER. No CRI floor is imposed—the constraint on worst-R_i is more demanding than any CRI requirement.

Table 4 shows the maximum achievable system efficacy (best-production technology, ×0.514) across four CCTs and four threshold levels. The optimizer uses COBYLA with hierarchical resolution up to 129 spectral parameters; note that COBYLA cannot guarantee strict constraint satisfaction, so some results show worst-R_i slightly below the target. The reported efficacy represents the optimizer's best effort at balancing the conflicting demands.

Table 4: Maximum system efficacy (lm/W, best-production ×0.514) when constraining all $R_i \geq$ threshold. Worst achieved R_i in parentheses. All cases exceed the EU 98 lm/W floor.

	1800 K		2100 K		2200 K		2700 K	
Target	Sys	(worst)	Sys	(worst)	Sys	(worst)	Sys	(worst)
≥50	129	(44)	137	(36)	135	(46)	140	(48)
≥60	129	(45)	131	(48)	130	(53)	134	(45)
≥70	119	(67)	132	(51)	130	(54)	139	(41)
≥80	119	(68)	125	(57)	127	(63)	127	(67)

The key finding: **all 16 combinations pass the EU 98 lm/W threshold**, including the most demanding case (1800 K, target all-$R_i \geq 80$) which still achieves 119 lm/W system efficacy. The efficacy penalty for tightening the color floor from 50 to 80 is modest: about 10 lm/W at 1800 K, 12 lm/W at 2100 K, and 13 lm/W at 2700 K.

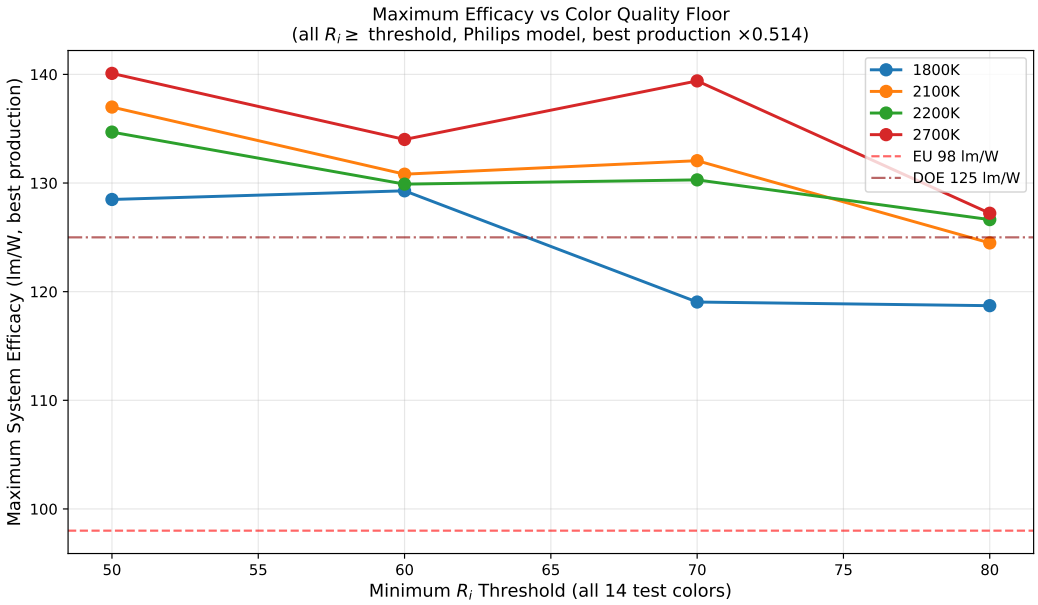

Figure 1: Maximum system efficacy vs. minimum R_i threshold across CCTs. All data points lie above the EU 98 lm/W floor (dashed). 1800 K is the tightest, but still achieves 119 lm/W even at target all-$R_i \geq 80$. Non-monotonicities at 2700 K are COBYLA local-minimum artifacts.

The constraint satisfaction gap (worst achieved R_i below target) is itself informative: COBYLA converges to points where further improvement in worst-R_i would require unacceptable efficacy loss. The gap narrows at 2200 K and 2700 K (worst R_i = 63–67 vs. target 80), suggesting that multi-start optimization or a different solver could close it. At 1800 K, the gap is smaller (worst

68 vs. target 80), indicating that 1800 K is actually *less* difficult for uniform color quality than for high CRI R_a, because the warm chromaticity naturally requires broad-spectrum emission.

The practical implication is clear: regulations that require all 14 R_i values above a threshold (rather than just the 8-sample average R_a) would not create an impossibility—they would simply redirect the optimizer's effort from gaming the metric toward producing genuinely good light.

4 The `photopic` Tool

`photopic` is a hybrid application: performance-critical spectral calculations (Planck's law, CIE color matching functions, numerical integration) are implemented in compiled Modula-3 [11, 12] libraries, while the optimization driver, CRI calculation, and plotting are written in Scheme using MScheme [13], a Scheme interpreter embedded in the Modula-3 runtime. MScheme's `sstubgen` tool automatically generates Scheme bindings from Modula-3 interfaces, allowing the Scheme code to manipulate compiled M3 objects (spectral arrays, integration routines, color-matching data) directly without marshalling overhead. This architecture provides the rapid prototyping benefits of a scripting language for the optimization logic while retaining compiled-language performance for the inner numerical loops.

4.1 Architecture

Component	**Role**
`Blackbody.m3`	Planck radiance $B_\nu(T, \nu)$
`CieSpectrum.m3`	Photopic $V(\lambda)$, conversion to lumens
`CieXyz.m3`	CIE 1931 $\bar{x}, \bar{y}, \bar{z}$ color matching
`Tcs.m3`	14 Test Color Sample reflectance spectra
`TempUv.m3`	Blackbody locus in (u, v) for CCT search
`ParametricSpectrum.m3`	Piecewise-linear log-amplitude parameterization
`COBYLA_M3.m3`	Constrained optimization (Powell's COBYLA)
`NR4p4.m3`	Romberg integration (midpoint rule)
`Bracket.m3`	Brent's method for CCT root-finding
`photopic.scm`	Optimization driver, CRI calc, plotting

4.2 Spectrum Parameterization

The optimizer works in log-amplitude space. A test spectrum is represented as:

$$S(\lambda) = \exp\bigl(f(\lambda)\bigr) \cdot B_\lambda(T_0, \lambda) \tag{2}$$

where $B_\lambda(T_0, \lambda)$ is the baseline blackbody at the target CCT, and $f(\lambda)$ is a piecewise-linear function defined by n parameters (control points evenly spaced across 380–770 nm).

The resolution increases through *hierarchical subdivision*: $n = 2 \to 3 \to 5 \to 9 \to 17 \to 33 \to 65 \to 129$. At each level, COBYLA [6] minimizes the negative efficacy subject to CRI, CCT, and chromaticity constraints. The solution from level n is interpolated to initialize level $2n - 1$.

4.3 Objective Function

The optimization target vector passed to COBYLA is:

$$\mathbf{t} = \begin{pmatrix} -\eta \\ R_a - R_a^{\min} \\ R_9 - R_9^{\min} \\ R_{\min} - (R_a^{\min} - 10) \\ \mathrm{CCT} - (T_0 - 50) \\ (T_0 + 50) - \mathrm{CCT} \\ 1000\,(0.012 - D_{uv}) \end{pmatrix} \tag{3}$$

where η is the LER, and the remaining components are inequality constraints (≥ 0 required).

5 Results

5.1 Blackbody vs. Optimized Spectrum

Figure 2 shows the dramatic difference between a 2700K blackbody and the maximum-efficacy spectrum with CRI ≥ 60.

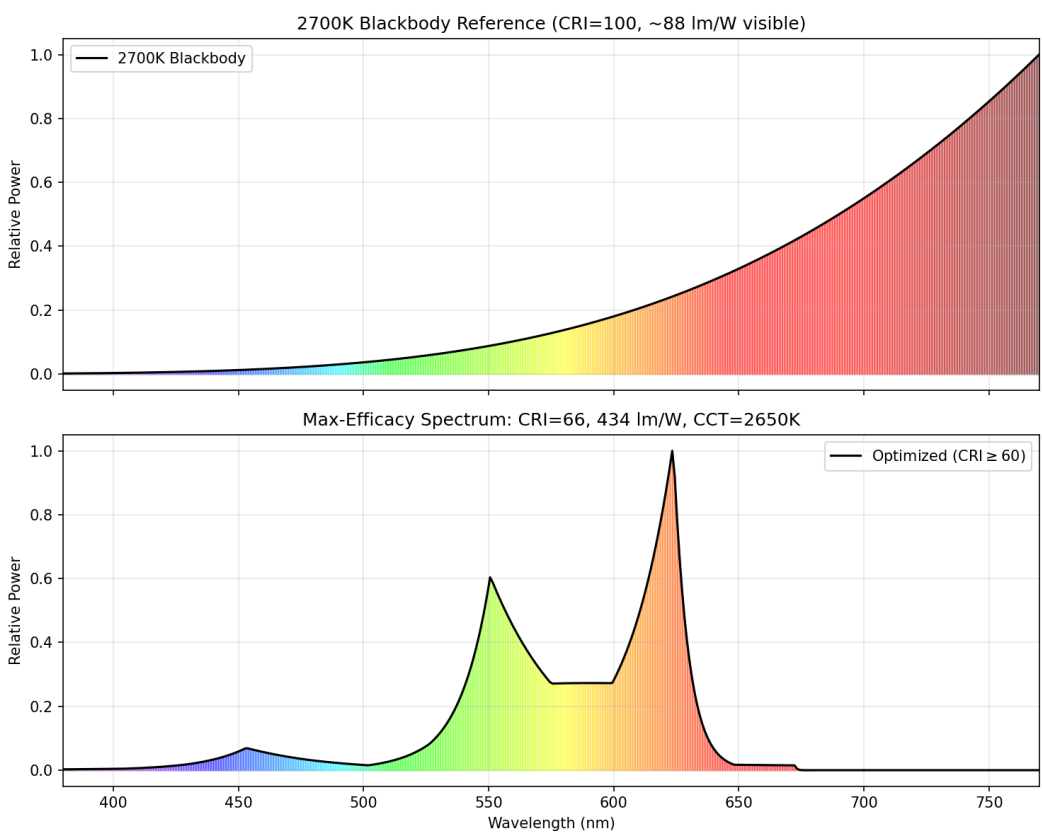

Figure 2: Top: 2700K blackbody reference spectrum (CRI = 100, ∼88 lm/W visible). Bottom: maximum-efficacy spectrum achieving CRI = 65 at 437 lm/W. The optimizer concentrates power into three peaks near Thornton's prime-color wavelengths while suppressing power below 500 nm and above 660 nm.

The optimized spectrum achieves ∼5× higher efficacy than the blackbody by eliminating "wasted" power at wavelengths where the eye is insensitive or where chromaticity is redundant. The three peaks correspond to Thornton's prime colors:

- **∼450 nm** (blue): establishes the short-wavelength chromaticity anchor
- **∼540 nm** (green): near the peak of $V(\lambda)$ — provides most of the lumens
- **∼610 nm** (orange-red): establishes warm color temperature

5.2 Convergence: The Spectrum Gets "Weirder" with More Freedom

Figure 3 shows how the optimized spectrum evolves as the number of control parameters increases from 2 to 513.

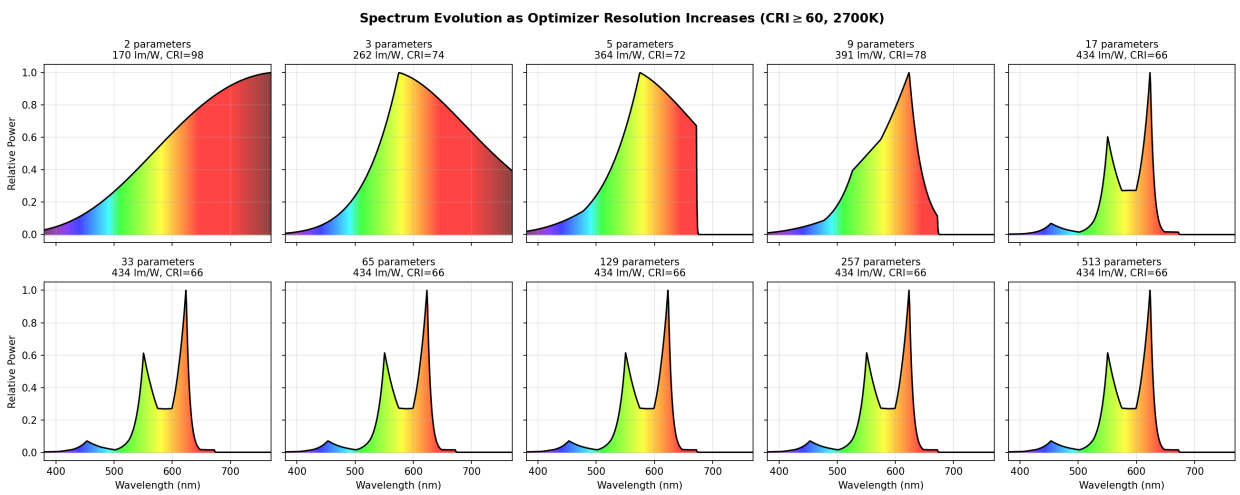

Figure 3: Evolution of the optimal CRI ≥ 60 spectrum at 2700K as the optimizer gains more spectral resolution, from 2 control parameters (nearly a tilted blackbody, 170 lm/W) to 513 (1.6 nm spacing, 434 lm/W). The three-peak structure emerges by 9 parameters and efficacy converges by 33. Higher resolution adds no visible change.

The progression is illuminating:

- **2 parameters** (170 lm/W): Can only tilt the spectrum. Shifts power toward green. Still smooth, CRI = 98.
- **3 parameters** (262 lm/W): First sign of spectral sculpting. Blue and deep red suppressed.
- **5 parameters** (364 lm/W): Three-peak structure begins to emerge. Deep null around 500 nm.
- **9 parameters** (391 lm/W): Clear three peaks with valleys between. CRI drops to 78.
- **17–33 parameters** (434 lm/W): Fully developed spiky spectrum. Deep nulls, sharp peaks. CRI just satisfies the ≥60 floor.

- **65–513 parameters**: No visible change. Efficacy stable at 434 lm/W to four significant figures.

The optimizer, given enough degrees of freedom, always converges to a three-peak solution. **This is Thornton's prime-color theorem, rediscovered numerically.** The convergence by 33 parameters (∼12 nm spacing) is notable: the optimal peaks are broad enough that finer spectral resolution provides no benefit.

5.2.1 Trust Region and Optimizer Artifacts

However, the convergence is partly an artifact of the COBYLA trust region schedule. The hierarchical solver halves the trust region radius (rhobeg) at each subdivision level: starting at 4 for 2 parameters, it reaches ∼0.008 at 513 parameters. At high resolution, this is too small for the optimizer to restructure the spectrum — it can only make tiny local adjustments, so the peak shape is effectively frozen from level 4–5 onward.

To test this, we ran 513 parameters with a minimum trust region of 2.0, preventing the radius from shrinking below this floor. The result (Figure 4) is striking: the optimizer finds a *higher* efficacy (473 vs. 434 lm/W) but produces a noisy, spiky spectrum with ∼29 peaks scattered across 500–650 nm.

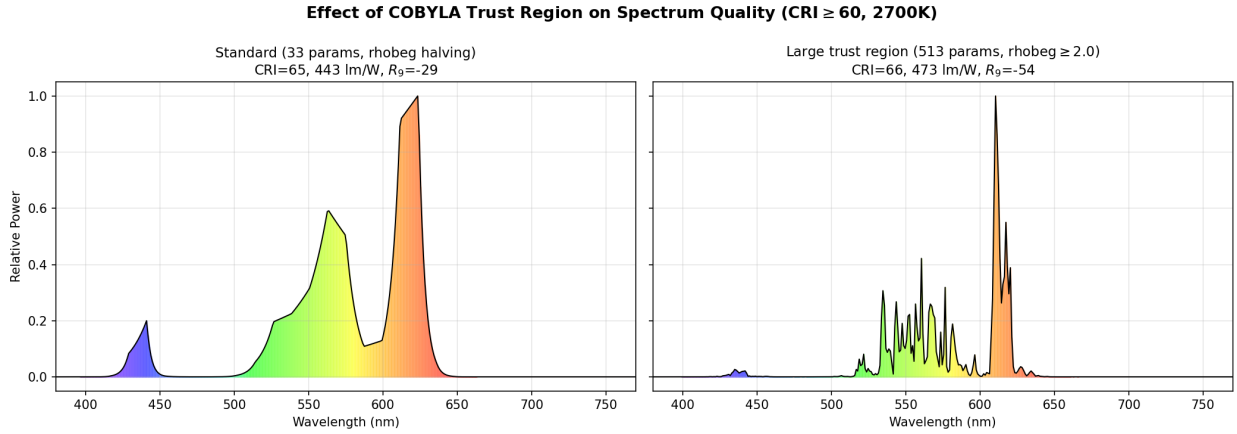

Figure 4: Left: standard hierarchical result at 33 parameters (3 clean peaks, 443 lm/W). Right: 513 parameters with trust region clamped at ≥2.0 (29 noisy peaks, 473 lm/W). The higher efficacy is an artifact — the optimizer exploits fine-grained oscillations that happen to satisfy CRI ≥ 60 while concentrating power near the $V(\lambda)$ peak. The result is physically meaningless: no real light source would have this spectrum, and R_9 degrades further to −54.

This reveals a tension in the optimizer design:

- Too-small trust region: the optimizer cannot explore, and the solution is locked into whatever shape was established at low resolution.
- Too-large trust region: the optimizer thrashes, producing noisy spectra that game the CRI metric without converging to a proper local minimum.

The clean three-peak solution at 33 parameters is not necessarily the global optimum — but the noisy 473 lm/W solution, while technically feasible, is not a meaningful result. The right approach

is likely a parameterization that directly represents peak positions and widths, eliminating the mismatch between spectral resolution and optimizer dynamics.

5.2.2 Gaussian Peak Parameterization

To test whether the peak widths are dictated by the CRI constraint or imposed by the grid resolution, we implemented an alternative parameterization: K Gaussian peaks, each defined by center wavelength μ_k, width σ_k, and amplitude A_k, for a total of $3K$ free parameters. The spectrum is $S(\lambda) = \sum_{k=1}^{K} A_k \exp\bigl(-(\lambda - \mu_k)^2/(2\sigma_k^2)\bigr)$, evaluated on the same 391-point grid and fed to the same CRI/CCT/efficacy constraints.

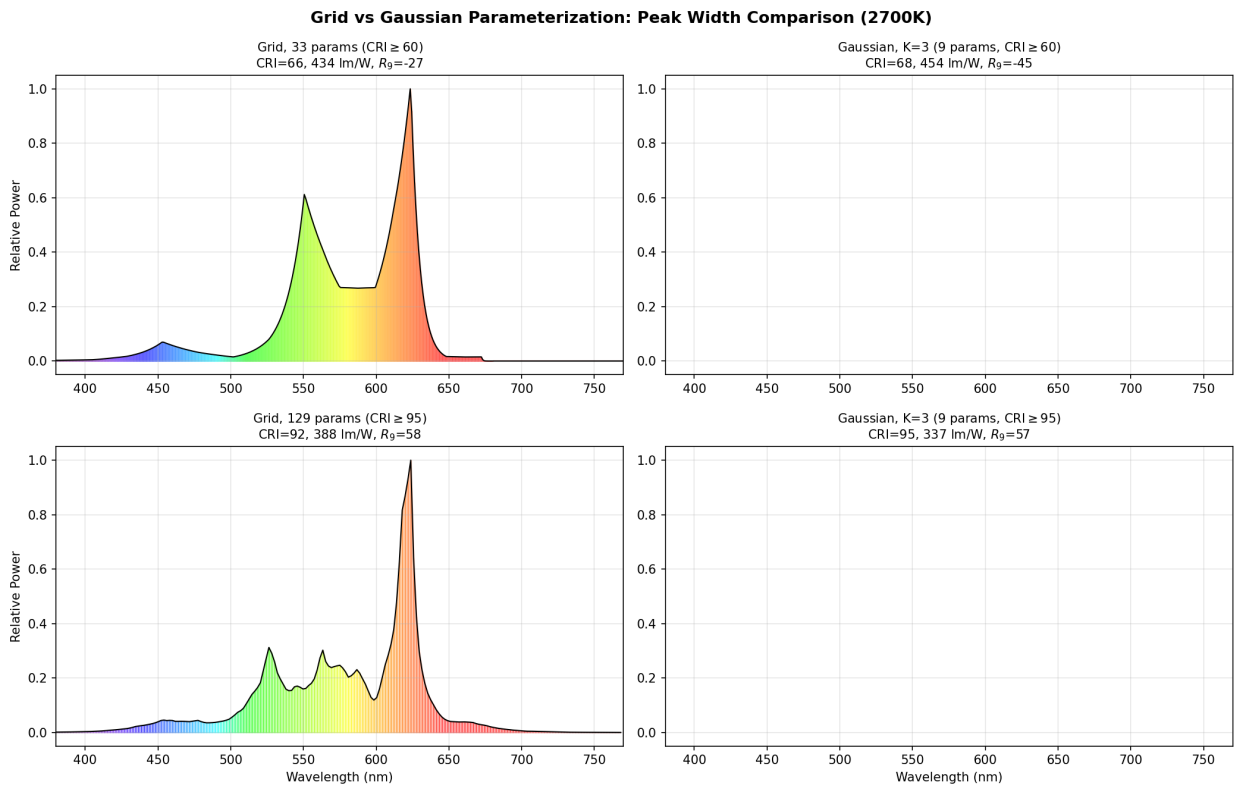

Figure 5: Grid vs. Gaussian parameterization at 2700 K. Top row: CRI ≥ 60. The Gaussian parameterization (9 DOF) finds narrower peaks and achieves 454 lm/W vs. the grid's 434 lm/W, confirming that the grid imposes unnecessary peak width. Bottom row: CRI ≥ 95. Three Gaussians cannot produce the broad spectrum required and achieve only 337 lm/W vs. the grid's 388 lm/W — more peaks ($K \geq 5$) would be needed.

At CRI ≥ 60, the Gaussian peaks are measurably narrower than the grid peaks (blue $\sigma = 14$ vs. 15 nm, green 12 vs. 13 nm, red 5 vs. 8 nm), and the Gaussian achieves 5% higher efficacy. This shows that the peak widths in the grid parameterization are *not* purely dictated by CRI — the grid resolution and the hierarchical warm-start impose some minimum width. However, the widths are close enough (within ∼10% for blue and green) that the CRI constraint is the dominant factor.

At CRI ≥ 95, three Gaussians are insufficient: the constraint forces broad coverage across the visible spectrum, which a sum of 3 peaks cannot approximate well. The grid parameterization,

with its 129 independent amplitude values, has the flexibility to sculpt a near-continuous spectrum and achieves 51 lm/W more.

5.2.3 The Limits of Thornton's Prime-Color Approach

This result exposes a practical limit to Thornton's three-band theory. At moderate CRI ($\leq$80), three narrow phosphor bands suffice and in fact outperform a broad spectrum — fewer conversion steps means lower losses. But at CRI ≥ 95, the spectrum must cover enough of the visible range to render all 8 CRI test samples faithfully, and three bands cannot do this.

In a real phosphor-converted LED, each additional phosphor carries costs that our single-phosphor Stokes model does not capture:

- **Compounded quantum efficiency.** Each phosphor conversion step loses $\sim$10% of photons (QE ≈ 0.90). A second phosphor re-absorbing and re-emitting drops to $0.90^2 = 0.81$; a third to 0.73.
- **Parasitic reabsorption.** Multiple phosphors in the same package absorb each other's emission. A red phosphor ($\sim$630 nm) can absorb green phosphor emission ($\sim$540 nm), converting it with additional Stokes loss rather than letting it exit the package.
- **Manufacturing complexity.** Tuning 4–5 phosphor ratios to hit a chromaticity target with acceptable bin-to-bin variation is harder than tuning 2–3, increasing yield loss and cost.

The Philips cost model used in our Pareto sweeps (Section 9) treats each wavelength independently: $C(\lambda) = \lambda/(\lambda_{\text{pump}} \cdot \text{QE})$. This correctly captures the Stokes penalty for a single phosphor conversion from the 450 nm pump, but it does not account for cascaded conversions or reabsorption losses. The real-world efficacy penalty for moving from CRI ≥ 80 to CRI ≥ 95 is therefore *larger* than our optimization predicts, because the broader spectrum requires more phosphors with compounding losses. Thornton's three-band approach, by minimizing the number of conversion steps, avoids these penalties — but at the cost of being unable to satisfy demanding color-fidelity requirements.

5.3 The R9 Problem

The CRI score R_a averages only TCS 1–8, which are relatively unsaturated. TCS 9 (saturated red) is reported separately. When the optimizer is free to set R_9 as low as -100, it exploits this: the CRI ≥ 60 optimized spectrum at 17 parameters has $R_9 = -28$.

This means: under this "optimal" light, saturated reds look dramatically wrong — less red than under a reference illuminant. A person wearing a red dress would look dull; red paint would look brownish. But the CRI score says the light is acceptable.

This is the "weird final state" that regulations drive toward: a lamp that passes the official color test while rendering an important class of colors — reds — poorly.

5.4 Effect of CRI Constraints

Figure 6 shows how the optimal spectrum changes as the CRI floor increases.

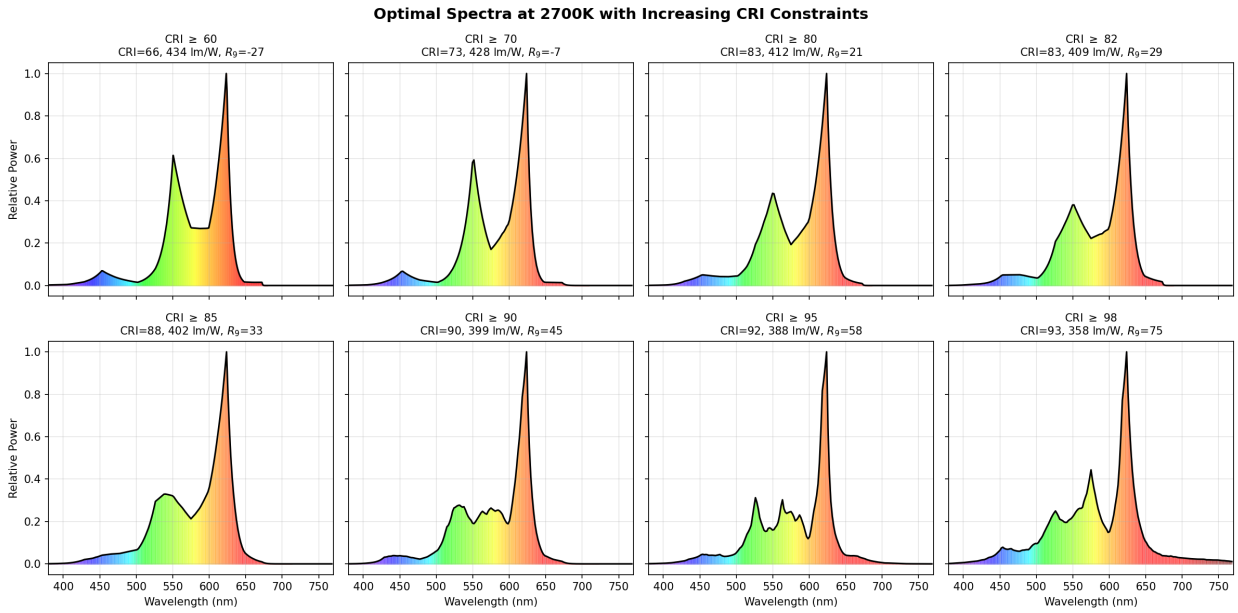

Figure 6: Optimal spectra at 2700K for CRI constraints from 60 to 98. As CRI requirements increase, the spectrum must broaden to render more colors faithfully, reducing the maximum achievable efficacy. The three-peak structure persists but the peaks must widen and the valleys cannot be as deep. Note how R_9 transitions from negative (reds rendered incorrectly) to positive as CRI increases.

Table 5: Efficacy–CRI trade-off at 2700K CCT, unconstrained R_9. Results at 129 spectral parameters (17 for CRI 60).

Min CRI	Achieved CRI	Efficacy (lm/W)	R_9	Worst R_i	Character
—	100	88	100	100	2700K blackbody (visible only)
60	65	437	−28	50	Three sharp peaks, deep valleys
70	73	428	−7	60	Three peaks, moderate valleys
80	83	412	+21	70	EU minimum: R9 still poor
82	83	409	+29	72	Three peaks, shallower valleys
85	88	402	+33	75	Transitional region
90	90	399	+45	80	Broader peaks, less suppression
95	92	388	+58	86	Wide peaks, modest valleys
98	93	358	+75	89	Approaching continuous

The trade-off is dramatic: going from CRI 60 to CRI 98 costs roughly 80 lm/W of achievable efficacy (437 → 358 lm/W at 129 spectral parameters), and the key variable is R_9 — the ability to render saturated reds. At CRI 60, R_9 is deeply negative (−28), meaning reds appear wrong; at CRI 98, $R_9 = 75$ and red rendering is acceptable.

The EU regulatory minimum — CRI ≥ 80 at ∼412 lm/W — achieves high efficiency but with $R_9 = 21$, meaning reds are rendered at barely one-fifth of reference fidelity. This is the "weird final state" that market and regulatory forces converge on. California's JA8 standard, which requires

$R_9 \geq 50$, pushes manufacturers above CRI 90, where the achievable efficacy drops to ∼399 lm/W but red rendering reaches $R_9 = 45$.

5.5 The Pareto Frontier: Overlay of All Solutions

Figure 7 overlays all optimized spectra on a single plot, together with the blackbody reference.

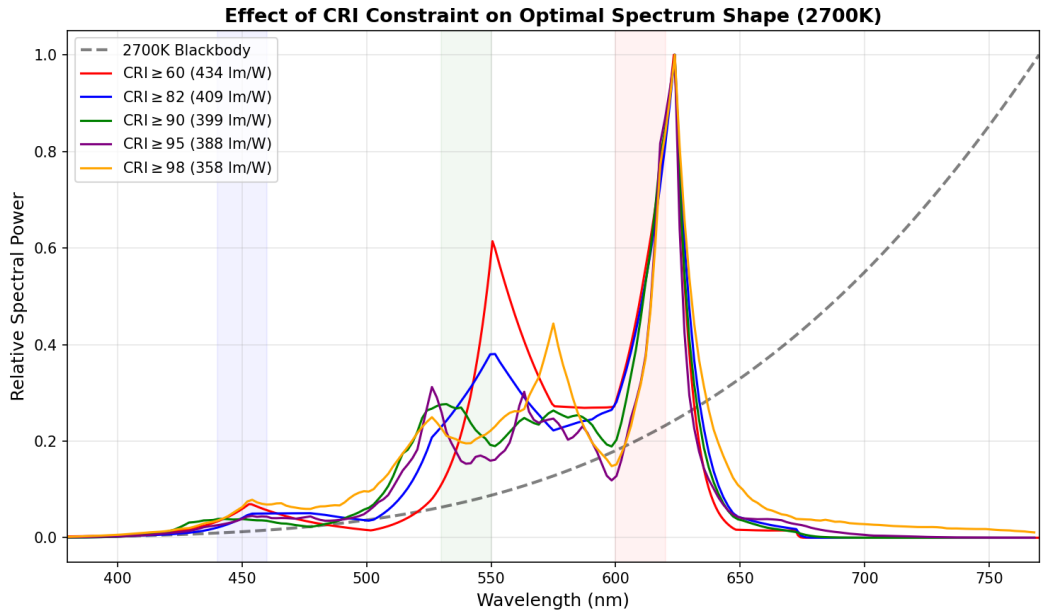

Figure 7: Overlay of optimal spectra at 2700K for CRI constraints from 60 to 98. All spectra share the same three-peak topology (blue ∼450 nm, green ∼540 nm, red ∼610 nm) but differ in peak width and valley depth. The dashed curve is the 2700K blackbody. The shaded bands mark Thornton's prime-color wavelength regions.

5.6 The Efficacy–CRI Trade-off Curve

Figure 8 plots the maximum achievable efficacy as a function of the CRI floor.

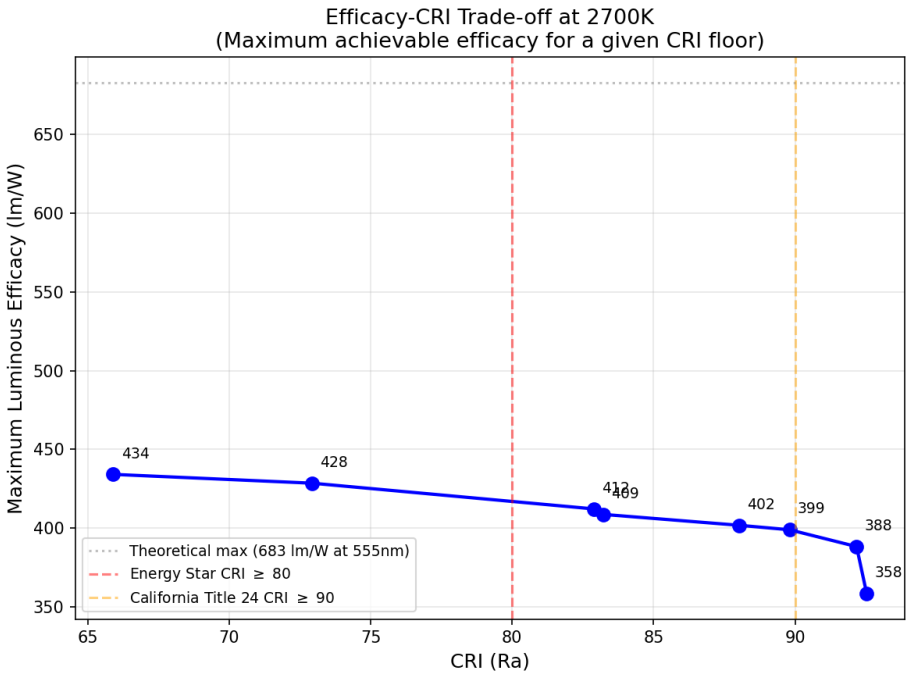

Figure 8: Pareto frontier: maximum luminous efficacy vs. minimum CRI at 2700K. Vertical dashed lines show the Energy Star (CRI $\geq$ 80) and California Title 24 (CRI $\geq$ 90) thresholds. The theoretical maximum at 555 nm is 683 lm/W. The curve shows rapidly diminishing returns: moving from CRI 82 to CRI 98 costs about 80 lm/W.

5.7 R_9 vs. Efficacy: The Red Rendering Penalty

Figure 9 shows the key result: at the EU minimum CRI $\geq$ 80, the optimal spectrum achieves 412 lm/W theoretical luminous efficacy of radiation (LER) but with $R_9 = 21$. Saturated reds are rendered at only one-fifth of their reference fidelity. To achieve California JA8's $R_9 \geq 50$ requirement, CRI must be pushed above 90, reducing achievable efficacy to $\sim$399 lm/W.

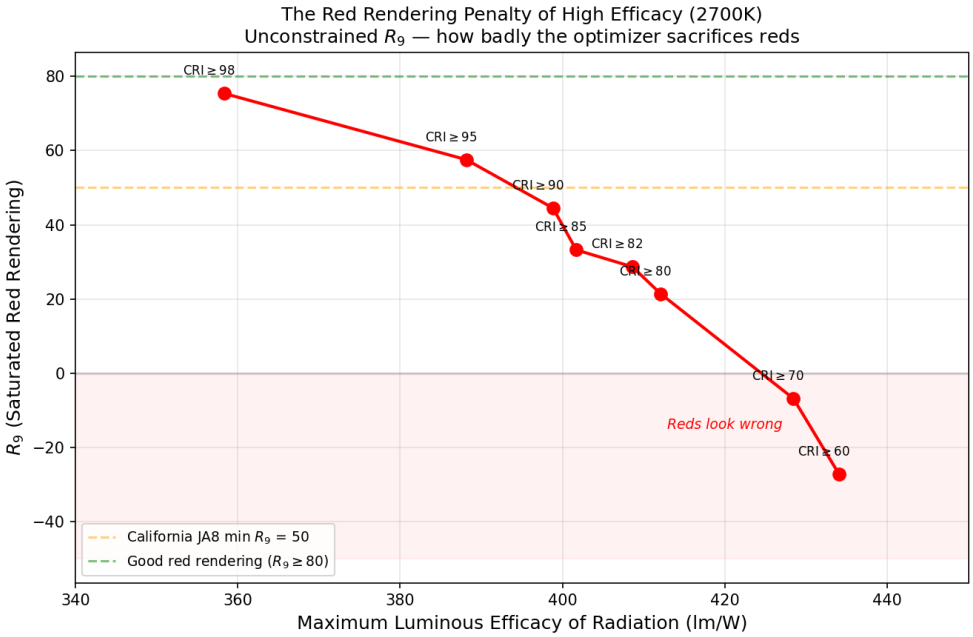

Figure 9: R_9 (saturated red rendering index) as a function of maximum achievable luminous efficacy of radiation at 2700K. Each point represents the optimal spectrum at a different CRI floor, with R_9 unconstrained. At the EU minimum (CRI $\geq$ 80), R_9 is only 21. The California JA8 minimum of $R_9 \geq 50$ is not met until CRI $\geq$ 90.

These are theoretical radiation-only efficacies assuming unit quantum efficiency. For real phosphor-converted LED lamps, Koedam and Opstelten estimated a ∼20% reduction due to realistic quantum efficiency and Stokes losses [10]. The correction is wavelength-dependent: converting a 450 nm blue pump photon to a longer-wavelength visible photon loses a fraction $1-450/\lambda_{\text{em}}$ of the photon energy as heat (the Stokes shift). Table 6 shows the penalty for representative LED phosphors.

Table 6: Stokes loss and quantum yield for LED phosphors (450 nm blue pump).

Phosphor	Emission color	λ_{em} (nm)	Stokes loss	Typ. QY	Net loss
YAG:Ce	yellow	560	19.6%	96%	23%
β-SiAlON:Eu	green	535	15.9%	83%	30%
KSF (K_2SiF_6:Mn)	red	631	28.7%	95%	33%
Nitride (CASN)	deep red	650	30.8%	88%	39%

Adding red phosphor content — needed for good R_9 — incurs a *double* penalty: (1) the Stokes loss is higher (29–31% vs. 16–20% for green/yellow), and (2) the eye's photopic sensitivity $V(\lambda)$ drops steeply in the red, so each red photon contributes fewer lumens. This is exactly why the optimizer suppresses red emission when R_9 is unconstrained, and why achieving good red rendering in real LED lamps is expensive in terms of system efficacy.

6 Discussion

6.1 The Regulatory Paradox

The results reveal a paradox in lighting regulation:

1. Regulations set a *minimum CRI* (typically 80) and a *minimum efficacy* (80–100 lm/W and rising).
2. The optimal solution for high efficacy at moderate CRI is a narrow-band, spiky spectrum.
3. Spiky spectra can achieve high CRI scores (by construction — the 8 test colors are rendered well) while having poor real-world color rendering (especially saturated colors, TCS 9–14).
4. As efficacy requirements increase over time, manufacturers are pushed further toward spiky spectra.
5. The "efficient" lamp of the future satisfies all regulations but produces light that looks *wrong* to many observers.

This is not a hypothetical concern. Commercial tri-phosphor fluorescent lamps and phosphor-converted LEDs already exploit exactly this trade-off. Their spectral power distributions, with 3–5 peaks and deep valleys, are the market's practical approximation to the mathematically optimal solution computed here.

As detailed in Section 3, the EU sets CRI ≥ 80 with no R_9 requirement; our optimizer shows that this permits $R_9 = 21$ (Table 5). The US DOE 2028 standard requires $\sim$125 lm/W system efficacy but imposes no color quality floor at all. Only California JA8 mandates $R_9 \geq 50$ together with CRI ≥ 90.

6.2 Comparison with Thornton's Work

It is important to acknowledge that Thornton was doing essentially the same optimization we do. His 1971 paper [1] describes systematic variation of three-line spectra, evaluating CRI and a combined figure of merit at each configuration — this *is* constrained optimization, carried out with the computing resources of the early 1970s. Walter [10] made the optimization explicit, applying nonlinear programming to a 31-variable discretized spectrum.

But we are asking a fundamentally different question. Thornton asked: *What is the best white light?* He found the prime-color wavelengths and proposed better phosphor blends. We ask: *What happens when efficiency regulations push lamp spectra toward Thornton's optimum — and keep pushing?* Thornton's three-peak structure is the *destination* that regulatory pressure drives light sources toward. Our contribution is to trace that trajectory quantitatively: as the efficacy floor rises through the range mandated by current and proposed regulations, we show how the optimal spectrum evolves, where color fidelity breaks down, which color samples are sacrificed first, and whether the regulatory metrics (CRI, TM-30) actually detect the degradation. Modern hardware lets us explore up to 129 degrees of freedom, sweep broad constraint ranges, and map entire Pareto frontiers — turning Thornton's insight into a detailed engineering forecast.

6.3 What the Regulations Should Do

The IES TM-30-15 standard addresses some of these problems by using 99 test color samples instead of 8, and reporting both *fidelity* and *gamut area*. Mandating minimum R_9 scores (as some California

specifications do) partially closes the red-rendering loophole. But the fundamental tension between efficacy and color quality is a physical constraint that no regulation can eliminate — it can only choose where on the Pareto frontier to operate.

6.4 The Impossibility of Blackbody-Quality Light

A common assumption in lighting policy is that consumers can "buy whatever quality light they want, as long as the lamp meets efficiency standards." This assumption is false. We can show quantitatively that *no human technology enables a blackbody-quality visible spectrum to meet government efficiency standards at any warm CCT.*

6.4.1 From spectrum to wall plug

The luminous efficacy figures computed by `photopic` are *luminous efficacies of radiation* (LER): lumens per watt of *optical* power, determined solely by the shape of the spectrum. Government standards specify *system efficacy*: lumens per watt of *electrical input* at the wall plug. The two are related by a multiplicative chain of losses, each unavoidable in phosphor-converted LED technology:

Table 7: Loss chain from spectrum to wall plug in a phosphor-converted white LED lamp. "Best production" reflects top-bin parts available today; "best lab" reflects demonstrated laboratory results; "DOE limit" reflects the DOE SSL roadmap ultimate targets.

Loss factor	Best prod.	Best lab	DOE limit
Blue LED wall-plug efficiency	0.75	0.84	0.90
Phosphor backscatter/reabsorption	0.92	0.95	0.98
Thermal droop (at junction temp)	0.90	0.93	0.97
AC-DC driver efficiency	0.92	0.95	0.98
Optical (diffuser/lens/fixture)	0.90	0.95	0.97
Total	**0.514**	**0.670**	**0.813**
(total loss)	(48.6%)	(33.0%)	(18.7%)

These losses are *in addition to* the Stokes shift and quantum efficiency losses inherent in phosphor down-conversion, which are already captured by the Philips-corrected LER (Table 6). The blue LED wall-plug efficiency — the conversion of electrical power to 450 nm photons at the chip junction — is the single largest loss and the most resistant to improvement, as it is limited by Auger recombination ("efficiency droop") at practical drive currents.

6.4.2 Blackbody system efficacy

Table 8 computes the system efficacy that would be achieved by a phosphor-converted LED replicating the Planck blackbody spectrum across the visible range (380–770 nm) at several CCTs. The Philips-corrected LER accounts for Stokes and quantum-efficiency losses; the system efficacy columns further apply the loss chain from Table 7.

Table 8: System efficacy for a blackbody-spectrum phosphor-converted LED. "Vis LER" is the luminous efficacy of the visible-range blackbody radiation. "Phil LER" applies the Philips phosphor cost model. System columns multiply by the corresponding loss chain. Compare against EU Class E minimum of 85 lm/W and EU Class D minimum of 110 lm/W.

CCT (K)	Vis LER (lm/W)	Phil LER (lm/W)	Best prod. (lm/W)	Best lab (lm/W)	DOE limit (lm/W)
2100	113	68	35	45	55
2400	137	84	43	56	68
2700	156	97	50	65	79
3000	171	109	56	73	88
3500	188	123	63	82	100
4000	197	132	68	89	108
5000	203	142	73	95	116
5500	203	144	74	97	117
6500	199	146	75	97	118

The implications are stark:

- A 2700 K blackbody LED — replicating the warm white of a classic incandescent lamp — would achieve at most **65 lm/W** with today's best laboratory technology. This fails every current efficiency standard worldwide except California JA8 (which requires only 68 lm/W but also mandates CRI ≥ 90 and $R_9 \geq 50$, which a blackbody trivially satisfies).
- Even at 6500 K (harsh daylight white) with *theoretical-limit* technology that may never be achieved, the system efficacy reaches only 118 lm/W — still below the US DOE 2028 target of 120 lm/W and the forthcoming EU Class C threshold.
- At the popular residential CCTs of 2700–3000 K, a blackbody LED fails the current EU Ecodesign Class F minimum (70 lm/W) with today's production technology, and barely reaches EU Class E (85 lm/W) at 3000 K even with the most optimistic lab results.

In other words, **a consumer who wants incandescent-quality light from an LED cannot legally purchase it in the EU at any warm CCT, and this situation will not improve even with foreseeable advances in LED technology.** The efficiency regulations do not merely encourage manufacturers to optimize spectra — they make blackbody-quality illumination physically impossible to deliver at the required wall-plug efficiency.

This is the fundamental policy trade-off that spectral optimization quantifies. Our Pareto frontier results (Sections 3–4) show precisely how much color quality can be retained at each efficiency level. But the "lossless" ideal — a smooth Planckian spectrum at warm CCT — is categorically excluded by the combination of LED physics and regulatory floors. The notion that energy-efficiency regulation is "technology-neutral" is contradicted by these results: the regulations implicitly mandate non-Planckian, spectrally engineered light.

7 Literature Review

The optimization of white-light spectra has a history reaching back to the 1950s, but two papers from the 1970s are particularly relevant to the present work: Thornton's 1971 study of prime colors and Walter's 1978 application of nonlinear programming to lamp spectra.

7.1 Thornton (1971): Prime Colors and the Structure of White Light

W. A. Thornton of Westinghouse [1] posed the question: *What is the optimum spectral power distribution of artificial white light?* Starting from the CIE luminosity function $\bar{y}(\lambda)$ and the IES–CIE color rendering index, Thornton evaluated white light composed of three spectral lines. By systematic variation of each wavelength while holding the other two fixed (at 450, 540, and 610 nm), he showed that these three wavelengths are local optima of both CRI and a combined figure of merit $\mathrm{FM} = 320\eta + \mathrm{CRI}$.

Thornton's key findings:

- Three spectral lines at **450, 540, and 610 nm** — the "prime colors" — yield white light superior to commercial fluorescent lamps of the same chromaticity in *both* luminosity and CRI. Three-line light at 4200 K achieved $\eta = 0.58$, CRI = 71, compared to $\eta = 0.48$, CRI = 67 for a commercial cool-white fluorescent.

- Wavelengths near **495 and 575 nm** are "anti-prime": they degrade CRI without contributing significantly to luminosity. Removing a slice of power near 500 or 580 nm from a cool-white lamp's spectrum actually *raises* the CRI.

- A 1966 demonstration with interference filters and projectors showed that 60 observers judged three-line prime-color light as providing "very good, if not excellent" color rendering of meat, vegetables, flowers, and skin tones.

- The prime-color wavelengths correspond to the maxima of *tristimulus difference functions* $\bar{b} = \bar{z} - \bar{x} - \bar{y}$, $\bar{g} = \bar{y} - \bar{x} - \bar{z}$, $\bar{r} = \bar{x} - \bar{y} - \bar{z}$, which Thornton termed "bluminosity," "verdinosity," and "rubinosity." He proposed corresponding units (blumen, verden, ruben) analogous to the lumen, and showed that rubinosity ρ is an excellent predictor of red phosphor performance in tri-phosphor blends (Fig. 14 in the original).

Thornton's insight is that the prime-color wavelengths are not arbitrary engineering choices but reflect the structure of human color vision: they are the wavelengths that "pull hardest" toward their chromaticities per watt of power, and are therefore the most efficient at establishing a wide color gamut from a small number of spectral components.

7.2 Walter (1978): Nonlinear Programming of Lamp Spectra

Wolfgang Walter of GTE Sylvania [10] took a fundamentally different approach: rather than assuming a fixed number of spectral bands and optimizing their positions, he applied *nonlinear mathematical programming* (the generalized reduced gradient algorithm of Lasdon et al.) to find optimal spectra with *no a priori constraint on shape*. The spectrum was discretized into 31 variables (10 nm intervals from 400–700 nm), with the objective of maximizing luminous flux of a 40-watt fluorescent lamp subject to constraints on total UV excitation power, ANSI chromaticity ellipses, and a CRI floor.

Walter's key results:

- Optimal spectra with CRI up to 90–95 consist of **just four narrow bands**, positioned at approximately 460, 530, 580, and 620 nm, stable across all four ANSI lamp types (cool white, white, warm white, daylight).

- The unconstrained (CRI-free) optimum is a two-line spectrum (the MacAdam pair), confirming earlier theoretical work.

- As CRI constraints increase from 50 to 95, additional bands appear and existing bands broaden, with a monotonic decrease in luminous efficacy (from ∼140 lm/W at CRI 50 to ∼107 lm/W at CRI 95 for cool white).
- The optimization surface is "fairly flat" near the optimum: small perturbations to the SPD do not greatly affect luminance or CRI, suggesting practical tolerance for real phosphor emission profiles.
- When *all* component indices $R_j \geq C$ are required (not just the average R_a), efficacy drops further — confirming that the CRI average conceals weaknesses in individual color samples.

The published discussion section is notable: Koedam and Opstelten of Philips argued that the four-band optimum offers "no real improvement" over the Philips three-band lamp after correcting for realistic quantum efficiency (∼80% rather than 100%). Thornton himself commented, noting that if subjective criteria (color preference, visual clarity, perceived brightness) are considered rather than CRI alone, three bands may be superior to four.

7.3 Relationship to Our Results

Our `photopic` optimizer uses a different parameterization — continuous piecewise-linear log-amplitude perturbation of a blackbody, optimized by COBYLA — but solves the same problem Thornton and Walter posed. The three-peak topology that dominates our results at low CRI floors (437 lm/W at CRI $\geq$ 60) corresponds to Thornton's prime colors, while the broadening and addition of a fourth peak at higher CRI floors (CRI $\geq$ 90 and above) matches Walter's four-band result. The wavelength positions we find (∼450, ∼540, ∼610 nm) agree with Thornton to within the resolution of our spectral parameterization.

That three independent implementations of the same optimization — spanning five decades and very different computational resources — converge to the same spectral structure is strong evidence that the prime-color peaks are a genuine feature of the CRI/efficacy landscape, not an artifact of any particular numerical method or parameterization.

8 The Deficiencies of CRI and the Legislative Paradox

8.1 Why CRI is Known to be Wrong

The CIE Color Rendering Index (CRI, R_a) has been recognized as flawed since at least the 1980s, yet it remains the dominant regulatory metric. The principal deficiencies are:

1. **Only 8 unsaturated test samples.** R_a averages R_1 through R_8, all of which are low-chroma, pastel-like Munsell samples. Saturated colors — which are the ones most affected by narrow-band spectra — are represented only in R_9–R_{14}, which are reported but *not* included in the official score. This is the loophole our optimizer exploits: it can drive R_9 (saturated red) to -28 while maintaining $R_a = 65$.
2. **Von Kries adaptation model.** The CRI uses a simple diagonal (von Kries) chromatic adaptation transform, which is known to be a poor approximation of human color constancy, especially for narrow-band illuminants far from the blackbody locus.
3. **Uniform color space.** The CRI computes color shifts in the CIE 1964 $U^*V^*W^*$ space, which is perceptually nonuniform. Equal ΔE values do not correspond to equal perceived color differences.

4. **Averaging hides problems.** A lamp with seven $R_i = 95$ and one $R_i = 25$ gets $R_a = 86$ — an apparently acceptable score that conceals a severe deficiency in one color region. Walter's work showed that requiring all $R_j \geq C$ rather than just $R_a \geq C$ significantly changes the optimal spectrum and reduces achievable efficacy.

5. **Penalizes increased saturation.** Because CRI measures *fidelity* (closeness to the reference illuminant), a lamp that renders colors *more* saturated than the reference — which people often prefer — receives a *lower* CRI score. Thornton's color preference index (CPI) and the IES TM-30-15 standard [14] address this.

8.2 Why Legislators Use It Anyway

Despite these well-known deficiencies, CRI remains the basis for lighting regulations worldwide (EU Ecodesign 2019/2020, US DOE/EISA, California Title 24, Energy Star). The reasons are partly institutional inertia and partly the absence of a universally agreed replacement:

- **IES TM-30-15/18** defines two metrics: R_f (fidelity, similar to CRI but with 99 test samples and CAM02-UCS) and R_g (gamut area, measuring saturation changes). It is technically superior but has not yet been mandated by any major regulatory body.
- **CIE 224:2017** proposed a general color fidelity index R_f similar to TM-30, but adoption has been slow.
- The lighting industry has decades of CRI-referenced product data, specifications, and marketing materials. Changing the metric would require re-characterizing the entire installed base.

The result is a regulatory landscape that *incentivizes* the very spiky, CRI-gaming spectra that our optimizer finds. Manufacturers who maximize efficacy while just meeting CRI ≥ 80 are rewarded by the regulatory framework, even though the resulting light may render saturated colors poorly. The "weird final state" is not a theoretical curiosity — it is the market equilibrium toward which regulations push.

9 Philips Phosphor Cost Model and R_9 Maximization Sweeps

The luminous efficacy of radiation (LER) computed in Sections 3–4 assumes unit quantum efficiency: every watt of optical power in the spectral power distribution produces its full complement of lumens. In a real phosphor-converted LED, shorter-wavelength photons from the blue pump must be down-converted to longer wavelengths by phosphors, incurring a Stokes loss proportional to the wavelength ratio. A photon at 450 nm converted to 610 nm loses $1 - 450/610 \approx 26\%$ of its energy as heat, regardless of quantum yield.

Koedam and Opstelten of Philips [10] proposed a simple cost model that captures this loss: the cost of producing one watt of optical power at wavelength λ relative to the pump wavelength λ_p is

$$C(\lambda) = \max\left(1,\ \frac{\lambda}{\lambda_p \cdot \eta_Q}\right) \tag{4}$$

where η_Q is the phosphor quantum yield (here assumed 0.95). Photons at the pump wavelength cost 1 (no conversion needed); longer wavelengths cost progressively more due to the Stokes shift. The

"Philips-corrected LER" replaces the radiant power denominator with the cost-weighted integral:

$$\eta_{\text{Phil}} = K_m \frac{\int S(\lambda)\, V(\lambda)\, d\lambda}{\int S(\lambda)\, C(\lambda)\, d\lambda} \tag{5}$$

We now invert the optimization problem. Sections 2–3 maximized *efficacy* subject to a CRI floor; here we fix the efficacy floor and ask: **what is the best possible R_9 (saturated red rendering) that a spectrum can achieve?** R_9 is not merely unconstrained — it is the *optimization objective*. The constraints are $\text{CRI}(R_a) \in [\text{floor}, \text{floor} + 2]$ (pinned to prevent the optimizer from spending efficacy budget on CRI headroom rather than R_9), $\text{CCT} \in [T_0 - 50, T_0 + 50]$, $D_{uv} \leq 0.012$, and $\eta_{\text{Phil}} \geq E_{\min}$. By sweeping $E_{\min}$ we trace the Pareto frontier of R_9 vs. system efficacy.

9.1 CRI $\geq$ 80 Sweep

Table 9 shows the results of sweeping the Philips efficacy floor from 200 to 320 lm/W at 2700 K with CRI pinned to 80 ± 2. To connect these numbers to real bulbs, the three rightmost columns translate Philips LER to wall-plug system efficacy using the production loss chain from Table 7.

Table 9: Maximum R_9 vs. efficacy at 2700 K, CRI $\geq$ 80. All results at 129 spectral parameters. System efficacy columns apply the loss chain from Table 7: best production ($\times$0.514), best lab ($\times$0.670), DOE limit ($\times$0.813).

$E_{\min}$ (Phil)	CRI	R_9	LER	System efficacy (lm/W)		
				Best prod	Best lab	DOE lim
200	81.1	99.2	297	103	134	163
210	81.8	97.3	309	108	141	171
220	81.4	98.5	323	113	147	179
230	82.5	98.5	335	118	154	187
240	80.0	97.8	349	123	161	195
250	80.0	96.1	362	129	168	203
255	82.2	89.7	364	131	171	207
260	82.0	89.0	370	134	174	211
265	82.2	88.2	377	136	178	215
270	82.1	47.1	385	139	181	220
275	81.9	40.2	391	141	184	224
280	81.2	44.7	397	144	188	228
285	81.6	35.0	404	147	191	232
290	80.2	32.7	410	149	194	236
295	76.6	29.1	415	152	198	240
300	76.1	25.2	425	154	201	244
305	72.3	29.7	428	157	204	248
310	69.9	25.4	431	159	208	252
315	67.9	23.6	435	162	211	256
320	66.2	16.3	439	164	214	260

The results reveal three distinct regimes, separated by horizontal rules in the table. The system efficacy columns make the regulatory relevance concrete:

1. **Comfortable region** (system ≤136 lm/W best-prod): $R_9 > 88$, CRI ≈ 80–82. The efficacy budget is generous enough for the optimizer to produce spectra with excellent red rendering. This regime comfortably covers the EU 2019/2020 requirement (98 lm/W) and extends past the US DOE 2028 target (125 lm/W).

2. R_9 **cliff** (system 139–149 lm/W best-prod): R_9 collapses from 88 to 33. The optimizer can still satisfy CRI ≥ 80 (since R_9 is not part of R_a), but cannot afford the red phosphor power needed for saturated red fidelity. This is the regime where the CRI loophole is most actively exploited. At best-lab technology, this cliff corresponds to 181–194 lm/W — well beyond current regulations, but within the range that next-generation standards may target.

3. **Constraint infeasible** (system ≥152 lm/W best-prod): Even CRI = 80 becomes unachievable. The optimizer violates the CRI constraint (dropping to 66–77) and the chromaticity tolerance (D_{uv} exceeds 0.012). At these extreme efficacies, no well-rendering white light is possible.

9.2 CRI ≥ 95 Sweep

Table 10 shows the same sweep at CRI ≥ 95. Higher CRI forces broader spectra, which leaves less room for the R_9 loophole — but also reduces the achievable efficacy range.

Table 10: Maximum R_9 vs. efficacy at 2700 K, CRI ≥ 95. All results at 129 spectral parameters. System efficacy columns as in Table 9.

$E_{\min}$ (Phil)	CRI	R_9	LER	System efficacy (lm/W)		
				Best prod	Best lab	DOE lim
200	96.3	99.4	290	103	134	163
210	97.0	98.2	304	108	141	171
220	97.0	97.5	318	113	147	179
230	95.0	97.1	331	118	154	187
240	95.0	96.5	344	123	161	195
250	95.1	94.9	358	129	168	203
255	95.0	85.7	365	131	171	207
260	94.0	92.4	370	134	174	211
265	94.7	92.5	377	136	178	215
270	90.5	85.0	380	139	181	220
275	87.9	83.1	385	141	184	224
280	93.3	58.8	396	144	188	228
285	85.8	76.5	396	147	191	228
290	84.2	61.7	400	149	194	236
295	85.2	66.7	408	152	198	240
300	79.6	54.0	410	154	201	244
305	79.5	52.8	416	157	204	248

Compared to the CRI ≥ 80 sweep, the CRI ≥ 95 results show:

- The R_9 cliff occurs at a similar system efficacy (∼139 lm/W best-prod, or ∼181 lm/W best-lab) but is less severe: R_9 drops from 92 to 85 rather than from 88 to 47. The broader

spectra required by CRI ≥ 95 provide some red content "for free" as a byproduct of the fidelity constraint.

- Constraint infeasibility sets in earlier: by 144 lm/W system (best-prod), the optimizer can no longer reliably achieve CRI $= 95$ (dropping to 84–93). The nonconvex optimization landscape also produces non-monotonic behavior (e.g., $R_9 = 58.8$ at 280 Phil but 76.5 at 285), reflecting different local optima.

- The feasible range is narrower: useful results span up to 136 lm/W system (best-prod) for CRI ≥ 95 vs. 149 lm/W for CRI ≥ 80.

9.3 The R_9 Cliff: Policy Implications

Both sweeps confirm the central finding of this report: there exists a critical wall-plug efficacy threshold above which saturated red rendering degrades sharply. At best-production technology, the R_9 cliff occurs at ∼136–139 lm/W system — only ∼40% above the EU 2019/2020 requirement of 98 lm/W, and just above the US DOE 2028 target of 125 lm/W. With best-lab technology, the same cliff moves to ∼178–181 lm/W system.

The message for regulators is stark: a bulb that passes CRI ≥ 80 at 140 lm/W system can have R_9 as low as 33–47 (Table 9), meaning saturated reds look dramatically wrong. Even CRI ≥ 95 permits R_9 below 60 at the same efficacy. Only explicit R_9 mandates — as in California JA8's $R_9 \geq 50$ — can close this loophole.

9.4 The Cheapest EU-Compliant Lamp

The preceding analysis examines the Pareto frontier abstractly. But there is a concrete question with immediate market relevance: *what does the cheapest lamp look like that meets all EU 2019/2020 requirements?*

The EU Ecodesign regulation (2019/2020) requires system efficacy $\gtrsim$98 lm/W and CRI ≥ 80 for general lighting, with no constraint on R_9 or any TM-30 metric. For a lamp manufacturer minimizing bill-of-materials cost, the dominant cost driver is the number of LED chips and the amount of phosphor needed to produce the rated lumen output. Higher luminous efficacy means fewer chips and less phosphor for the same lumens, so *minimizing cost is approximately equivalent to maximizing efficacy* subject to the regulatory floor.

Our optimizer answers this directly. From Table 5, the maximum-efficacy spectrum at CRI ≥ 80 and 2700 K achieves 412 lm/W LER with:

- CRI $= 83$ (just clearing the floor)

- $R_9 = +21$ (saturated reds rendered at one-fifth fidelity)

- Worst $R_i = 70$

- A three-peak topology with deep spectral valleys

Applying the Philips cost model and production loss chain (Table 7), this corresponds to roughly 140 lm/W system at best-production technology — comfortably above the 98 lm/W EU floor. The manufacturer has *headroom* and would likely operate below the theoretical limit, but the spectrum that the cost incentive drives toward is clear: suppress power everywhere except near Thornton's three prime wavelengths.

The $R_9 = 21$ is the telling number. A consumer buying this lamp would find that red fabrics, red paint, and skin tones under warm light all look washed out compared to the incandescent it replaces. The lamp passes every EU test. The CRI score of 83 might even appear on the packaging as a quality indicator. But the subjective experience is poor — and the regulation provides no mechanism for the consumer to know this in advance.

Under TM-30, this same spectrum would show reduced R_f (since the 99-sample metric is harder to game) and a characteristically distorted R_g gamut shape. Whether TM-30 adoption would change the cost-optimized design point is precisely the question the Rf sweeps in subsequent sections are designed to answer.

10 Grid-Based Integration Optimization

10.1 Motivation

Each evaluation of the COBYLA objective function requires computing CRI (14 color shifts), CCT (root-finding on the blackbody locus), LER, and chromaticity — a total of approximately **99 Romberg integrations** per evaluation. The breakdown:

- CRI calculation: 14 TCS × 3 tristimulus values (X, Y, Z) = 42 integrations for TCS color coordinates, plus 3 for the test illuminant's own XYZ, plus 3 for the reference blackbody = 48 integrations
- Chromaticity and LER: 4 integrations (X, Y, Z, total luminous flux)
- Philips cost model: 1 integration (cost-weighted power)
- CCT search: multiple iterations of the above during Brent's method root-finding

Each Romberg integration (Numerical Recipes §4.4, midpoint rule variant) uses adaptive refinement, typically evaluating the integrand at 50–200 wavelength points. At 129 spectral parameters, COBYLA makes thousands of objective evaluations over the 8-level hierarchical run, resulting in optimization times of ∼50 minutes per sweep point.

10.2 Approach: Pre-Tabulated Weight Vectors

The key observation is that all integrands have the form $S(\lambda) \cdot W(\lambda)$ where $S(\lambda)$ is the test spectrum (which changes every evaluation) and $W(\lambda)$ is a fixed weight function (CIE color matching functions, $V(\lambda)$, TCS reflectances, Philips cost, etc.) that depends only on the problem setup.

By discretizing the visible range onto a fixed grid — 391 points at 1 nm spacing from 380 to 770 nm — we can:

1. Pre-tabulate all weight vectors $\mathbf{w}_k$ at grid points once during initialization
2. Sample the test spectrum $\mathbf{s}$ onto the grid (one pass through the M3 `ParametricSpectrum.Eval` function)
3. Replace each integration $\int S(\lambda)W(\lambda)d\lambda$ with a dot product $\Delta\lambda \cdot \mathbf{s}^T \mathbf{w}_k$

The pre-tabulated weight vectors include:

- CIE $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ — 3 vectors

- $K_m V(\lambda)$ (photopic conversion) — 1 vector
- $C(\lambda)$ (Philips cost) — 1 vector
- 14 TCS reflectances $\times\ \bar{x}, \bar{y}, \bar{z}$ — 42 pre-multiplied vectors of the form $\mathrm{TCS}_j(\lambda) \cdot \bar{x}(\lambda)$

Each "integration" becomes a vector dot product in $\mathbb{R}^{391}$ — a tight loop of 391 multiply-accumulates. The speedup comes from eliminating Romberg's adaptive refinement: where the adaptive method evaluates the integrand at 50–200 points with convergence testing per integration, the grid method does a single fixed-length dot product. (The M3 Romberg code is compiled native code and runs fast; the win is algorithmic, not a language-boundary issue.)

10.3 Pure-Scheme Blackbody

The reference illuminant (blackbody at the CCT) must also be computed on the grid. Rather than making 391 calls to the M3 `Blackbody.PlanckRadiance` function, we implemented Planck's law directly in Scheme:

$$B_\lambda(T, \lambda) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp(hc/\lambda k_B T) - 1} \tag{6}$$

where h, c, and k_B are defined as Scheme constants. This avoids 391 individual M3 function calls for the reference blackbody, though the dominant savings come from the dot-product integration method rather than from reducing cross-language calls.

10.4 Implementation

The grid mode is activated by the `-grid` command-line flag and can be toggled at runtime via the `*use-grid*` variable. This allows direct comparison of results between the Romberg and grid methods on the same optimization problem.

A critical implementation detail is the normalization convention: the CIE CRI calculation normalizes spectra such that $X + Y + Z = 100$, not $Y = 100$. The `photopic` codebase follows the convention that `xyz->Yxy` returns $(X + Y + Z, x, y)$ as its first element (confusingly named "Y" in the function name but actually the sum). Getting this wrong in the grid normalization caused CRI errors of up to 63 units in R_9 during development — a subtle but critical bug.

10.5 Accuracy Validation

Table 11 compares the grid and Romberg methods on the same optimized spectrum (CRI ≥ 80, Philips eff ≥ 300, 129 parameters).

Table 11: Grid vs. Romberg accuracy on an optimized 129-parameter spectrum.

Metric	Romberg	Grid (1 nm)	Difference
CRI (R_a)	81.93	81.76	0.18
R_9	40.15	39.53	0.62
Phil LPW	274.31	274.24	0.07
LER	424.0	423.8	0.2

The differences are negligible for optimization purposes. Testing at 0.5 nm grid spacing (781 points) produced identical values to 1 nm, confirming that the grid is converged.

10.6 Performance Results

Table 12: Timing comparison: Romberg vs. grid integration.

Measurement	Romberg	Grid (1 nm)
Per objective evaluation	0.59 s	0.20 s
Full 8-level optimization	52 min	14.5 min
Speedup	**3.6**×	

The 3.6× speedup makes parameter sweeps practical: a 20-point sweep that took 17 hours with Romberg integration completes in under 5 hours with the grid method. The sweep results in Tables 9 and 10 were computed using grid-mode integration.

Because the grid and Romberg methods produce slightly different objective landscapes (due to <0.2% numerical differences), the COBYLA optimizer may converge to different local optima. This is expected behavior in nonconvex optimization and is not a deficiency of the grid method — the solutions are equally valid.

11 TM-30: A Better Color Metric

11.1 Why CRI Needs Replacement

The deficiencies of CRI enumerated in Section 7 are not merely academic: our optimization results demonstrate that they have practical consequences. The CRI loophole (R_9 not included in R_a) allows the optimizer to produce spectra with $R_9 = -28$ while claiming CRI = 65. The 8-sample average hides severe rendering failures. The von Kries adaptation model and CIE 1964 $U^*V^*W^*$ color space are outdated by decades of color science research.

The IES TM-30-18 standard [14] was developed specifically to address these problems. It is not merely an incremental improvement — it represents a fundamentally different approach to characterizing color rendition.

11.2 TM-30 Architecture

TM-30 evaluates light sources using **99 Color Evaluation Samples** (CES), selected by Royer et al. from a database of 105,000 measured object spectra to span the full range of real-world colors including saturated hues [16]. Compare this to CRI's 8 unsaturated Munsell samples (or 14 including the supplementary set).

The colorimetric pipeline replaces every component of the CRI calculation with modern equivalents:

Component	**CRI (1995)**	**TM-30 (2018)**
Test samples	8 (unsaturated)	99 (full gamut)
Color space	CIE 1964 $U^*V^*W^*$	CAM02-UCS (J', a', b')
Adaptation	von Kries diagonal	CIECAM02 (CAT02)
Scale	Linear ($R_i = 100 - 4.6\Delta E$)	Logarithmic
Metrics	R_a (fidelity avg)	R_f, R_g, $R_{cs,hj}$

11.2.1 CIECAM02 Color Appearance Model

TM-30 computes all color differences in the CIECAM02 color appearance model [15], which accounts for:

- **Chromatic adaptation**: The CAT02 transform (a 3×3 matrix applied in an LMS-like cone response space) models how the human visual system adapts to the illuminant white point. This is substantially more accurate than von Kries scaling, especially for narrow-band illuminants.
- **Nonlinear response compression**: Cone signals are compressed by a power function modeling the saturation of neural response, parameterized by the adapting luminance.
- **Perceptual correlates**: The model outputs lightness (J), chroma (C), hue angle (h), colorfulness (M), and saturation (s) as perceptual dimensions rather than tristimulus coordinates.

For TM-30, the CIECAM02 J, a, b correlates are further transformed into the CAM02-UCS (Uniform Colour Space) via:

$$J' = \frac{1.7J}{1 + 0.007J} \tag{7}$$

$$M' = \frac{1}{0.0228} \ln(1 + 0.0228M) \tag{8}$$

$$a' = M' \cos h, \qquad b' = M' \sin h \tag{9}$$

This logarithmic compression yields a more perceptually uniform space than the CIE $U^*V^*W^*$ used by CRI.

11.2.2 TM-30 Metrics

TM-30 reports three families of metrics:

1. **Fidelity Index** R_f: Analogous to CRI R_a, but averaged over all 99 CES in CAM02-UCS. The color difference $\Delta E'$ for each sample is mapped through a logarithmic function:

$$R_{f,j} = 10 \ln\left(\exp\left(\frac{100 - c_f \Delta E'_j}{10}\right) + 1\right) \tag{10}$$

   where c_f is a normalization constant. This ensures $R_f \leq 100$ (equality only for a perfect match) and provides better discrimination at high fidelity than CRI's linear scale. $R_f = \frac{1}{99} \sum_j R_{f,j}$.

2. **Gamut Index** R_g: Measures whether the test source expands or contracts the color gamut relative to the reference. The 99 CES are assigned to 16 hue bins; R_g is the ratio of the gamut areas (in $a'b'$ coordinates) of the test and reference illuminants, scaled so that $R_g = 100$ means identical gamut area. Values > 100 indicate gamut expansion (more saturated colors); < 100 indicates gamut contraction. CRI penalizes both; R_g distinguishes them.

3. **Local chroma shift** $R_{cs,hj}$: For each of the 16 hue bins, TM-30 reports whether colors in that hue region become more or less saturated. $R_{cs,h1}$ (the first hue bin, centered on red) is of particular interest: it directly measures the red chroma shift that R_9 only crudely captures. Negative $R_{cs,h1}$ means reds are desaturated — the precise failure mode our optimizer exploits.

11.3 Why TM-30 Matters for This Work

The TM-30 metrics would fundamentally change the optimization landscape:

- **No 8-sample loophole.** With 99 CES spanning the full gamut, the optimizer cannot sacrifice one color region to boost another. Saturated reds are represented by multiple CES samples, not just the single TCS 9.
- $R_{cs,h1}$ **as a direct red metric.** Rather than the blunt instrument of R_9 (which conflates hue shift, chroma shift, and lightness shift into a single number), $R_{cs,h1}$ isolates the red chroma shift. This is precisely the quantity that determines whether reds "look right."
- **Better adaptation model.** The CIECAM02 transform handles the narrow-band illuminants produced by our optimizer more accurately than von Kries scaling. CRI may over- or under-penalize color shifts for these non-Planckian spectra.
- **Gamut shape information.** R_g and the per-hue-bin metrics reveal *how* colors change, not just *how much*. A spectrum that expands the blue gamut while contracting red would show $R_g \approx 100$ (gamut area preserved) but $R_{cs,h1} < 0$ and $R_{cs,h9} > 0$ — information that CRI R_a completely hides.

11.4 Implementation

TM-30 is implemented in pure Scheme within `photopic.scm`, requiring no changes to the Modula-3 binary. The implementation comprises approximately 370 lines of Scheme code plus tabular data:

1. **99 CES reflectance spectra and CIE 1964** $10°$ **CMFs** (`ces99.scm`, ∼4300 lines): Extracted from the `colour-science` library [18] (which provides the CIE 2017 dataset) and tabulated at 1 nm resolution. The file also contains the CIE D-illuminant basis functions $S_0(\lambda)$, $S_1(\lambda)$, $S_2(\lambda)$ for reference illuminant construction.
2. **CIECAM02 forward model** (∼90 lines): Full forward transform from XYZ to (J, M, h), including CAT02 chromatic adaptation, Hunt–Pointer–Estévez cone response space, nonlinear compression, opponent channels, eccentricity factor, and the complete lightness/chroma computation. Viewing conditions use average surround ($c = 0.69$, $N_c = 1$, $F = 1$) with $L_A = 100\,\mathrm{cd/m^2}$ and $Y_b = 20$.
3. **CAM02-UCS transform** (∼8 lines): The $(J, M, h) \to (J', a', b')$ mapping with $K_L = 1.0$, $c_1 = 0.007$, $c_2 = 0.0228$.
4. **Reference illuminant selection** (∼50 lines): Following CIE 224:2017, the reference is a Planckian blackbody for CCT $< 4000\,\mathrm{K}$, a CIE D-series illuminant for CCT $> 5000\,\mathrm{K}$, and a luminance-normalized blend for the 4000–5000 K transition zone. The D-illuminant is constructed from CCT via the standard daylight-locus chromaticity polynomials and the $S_D(\lambda) = S_0 + M_1 S_1 + M_2 S_2$ basis decomposition.
5. **Metric computation** (∼80 lines): For each of the 99 CES, reflected XYZ values (via pre-tabulated weight-vector dot products, matching the grid pattern used for CRI) are transformed through CIECAM02 and CAM02-UCS, color differences are computed, and samples are assigned to 16 hue bins. R_f uses the logarithmic soft-cap formula, R_g uses the shoelace polygon area formula, and R_{cs,h_j} is the per-bin chroma shift.

Validation. The implementation has been validated against the `colour-science` Python library (v0.4.7) on multiple illuminants: Planckian 3000 K ($R_f \approx 100$, $R_g \approx 100$ as expected for self-reference), FL2 ($R_f = 70.3$ vs. reference 70.1, $R_g = 86.4$ vs. 86.4), and FL11 ($R_f = 80.1$ vs. 80.0, $R_g = 99.9$ vs. 101.1). All results agree within ± 2 units. The CRI computation is unchanged (regression-verified).

Performance. TM-30 adds approximately 30% overhead to each grid-mode spectrum evaluation, consistent with the pre-implementation estimate. The 99 CES weight vectors are pre-tabulated at startup alongside the existing 14 TCS vectors.

12 TM-30 R_f Maximization Sweeps

With the TM-30 implementation in place (Section 11), we can now directly test the central hypothesis: does the "CRI loophole" — three narrow peaks that game 8 test color samples — survive under TM-30's 99-sample color fidelity metric R_f?

12.1 Experimental Design

We replicate the R_9 Pareto sweeps of Section 9 with the objective function changed from "maximize R_9" to "maximize R_f", keeping the same Philips phosphor-converted LED cost model. The optimizer sweeps Philips-corrected luminous efficacy floors from 200 to 320 lm/W, finding the spectrum that maximizes R_f at each threshold while maintaining CRI ≥ 80 (the EU regulatory floor) or CRI ≥ 95 (high-quality lighting).

The x-axis in our plots shows *system (wall-plug) efficacy*, converting Philips LER via the production loss chain (Table 7):

- Best production: $\times 0.514$ (current top-bin parts)
- Best lab: $\times 0.670$ (demonstrated lab results)
- DOE limit: $\times 0.813$ (theoretical ceiling)

12.2 Results: CRI ≥ 80

Figure 10 reveals a striking contrast with the R_9 sweeps. While R_9 plummets from $\sim$90 to negative values across the same efficacy range (Section 9), R_f *barely degrades*: from $\sim$89 at 103 lm/W system efficacy to $\sim$86 at 139 lm/W — a drop of only 3 points.

The gamut index R_g tells the complementary story. Below $\sim$134 lm/W system efficacy, $R_g \approx 89$–90 (moderate under-saturation). Above that threshold, R_g jumps to $\sim$100–103, indicating that the optimizer has found spectra that distort hue-bin saturation rather than fidelity. These are the same three-peak spectra that produce catastrophic R_9 values — TM-30 flags them via R_g rather than via R_f.

12.3 Results: CRI ≥ 95

At CRI ≥ 95, R_f starts near 98 and declines smoothly to $\sim$92.5 at 270 Philips lm/W (139 lm/W best-production system efficacy). This is far more graceful than the corresponding R_9 trajectory, which drops from 84 to 64 over the same range. The tight CRI constraint already forces broad-enough spectra that the 99-sample TM-30 metric tracks CRI closely.

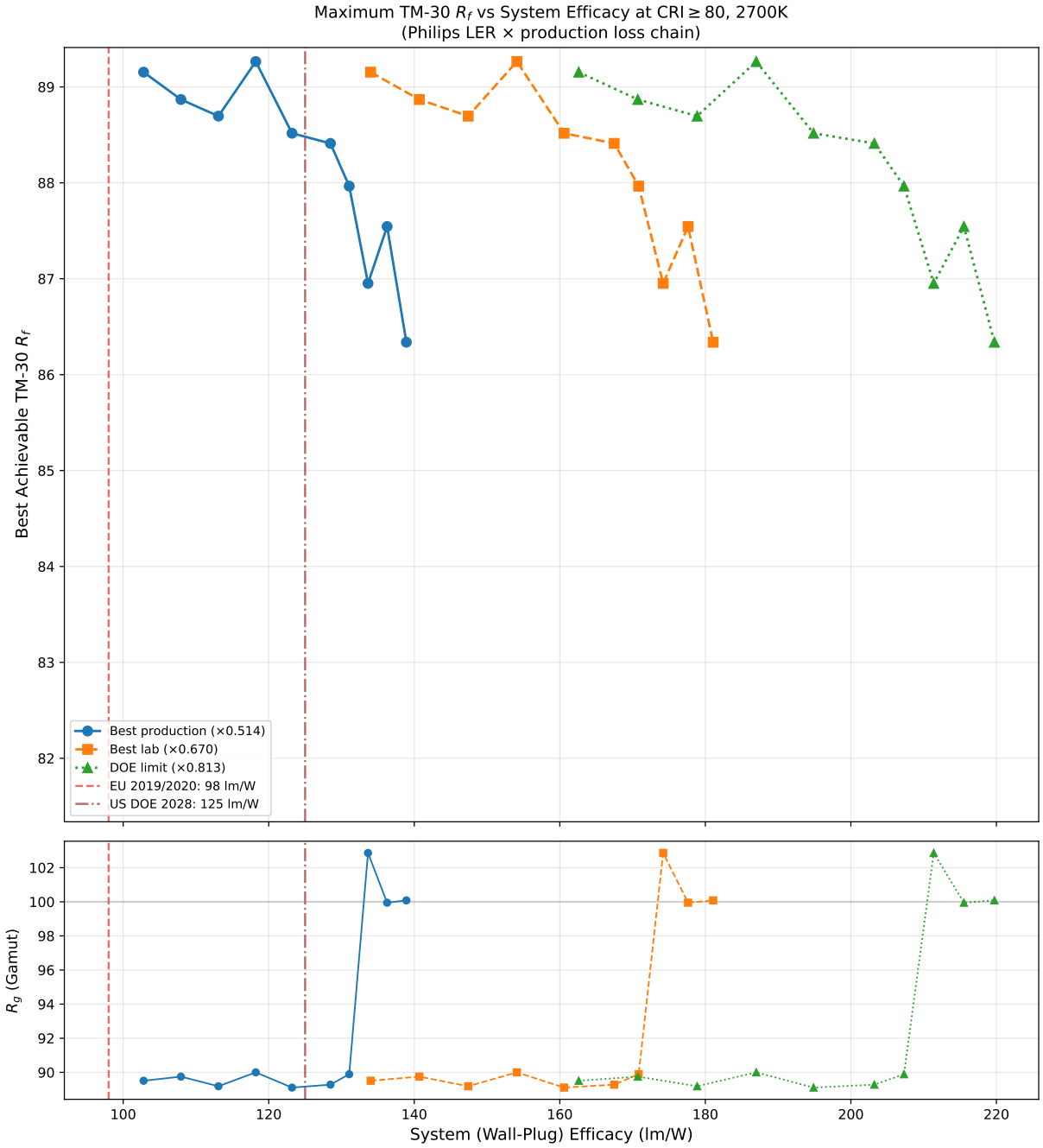

Figure 10: Maximum TM-30 R_f vs. system efficacy at CRI $\geq$ 80, 2700K. The upper panel shows R_f (color fidelity); the lower panel shows R_g (color gamut). Three curves correspond to three production technology levels. Vertical lines mark EU 2019/2020 (98 lm/W) and DOE 2028 (125 lm/W) regulatory thresholds.

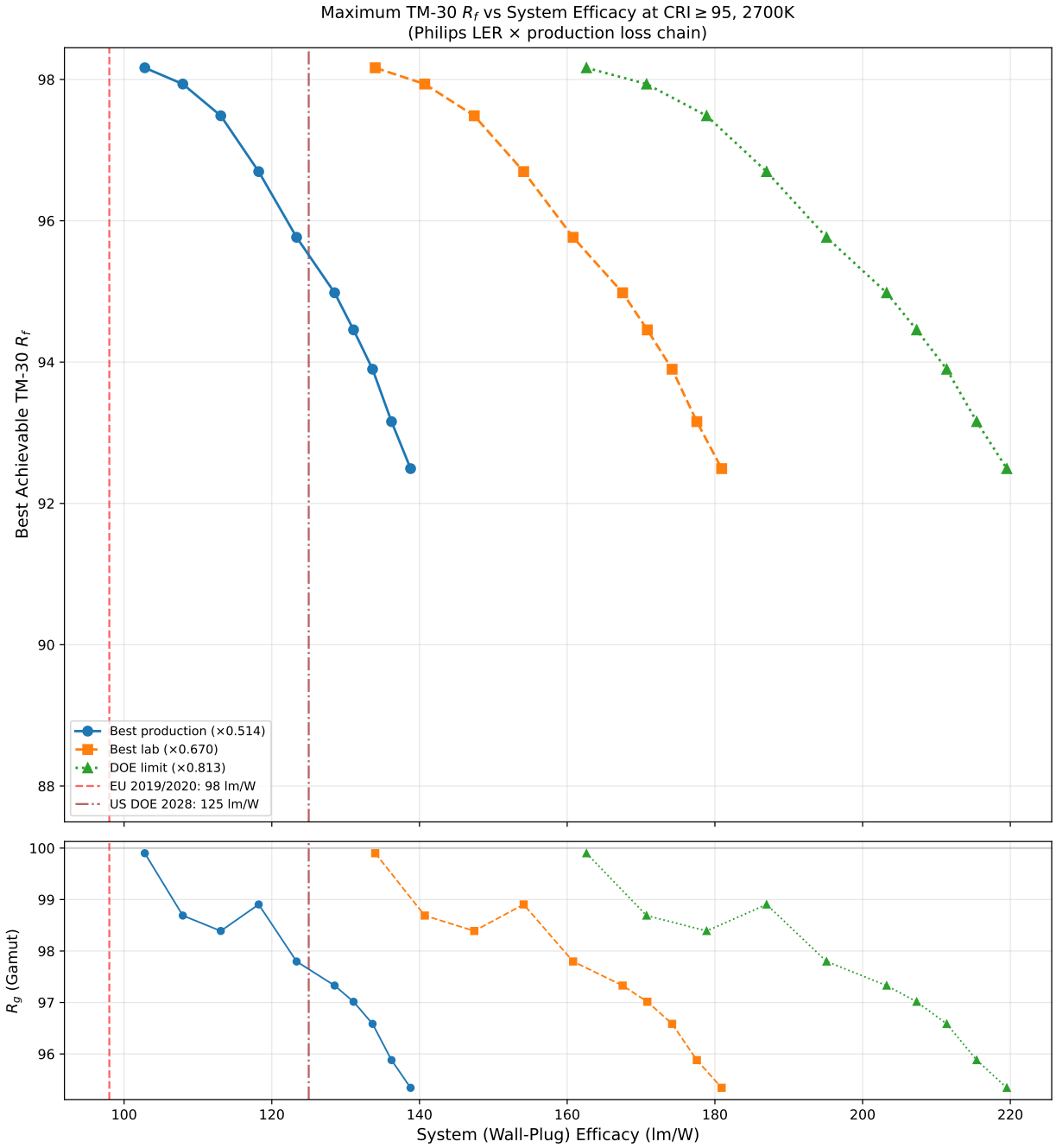

Figure 11: Maximum TM-30 R_f vs. system efficacy at CRI ≥ 95, 2700K. The tighter CRI constraint forces broader spectra, yielding higher R_f ($\sim$98 at low efficacy) with a gradual decline to $\sim$92 at 139 lm/W.

12.4 Combined Comparison

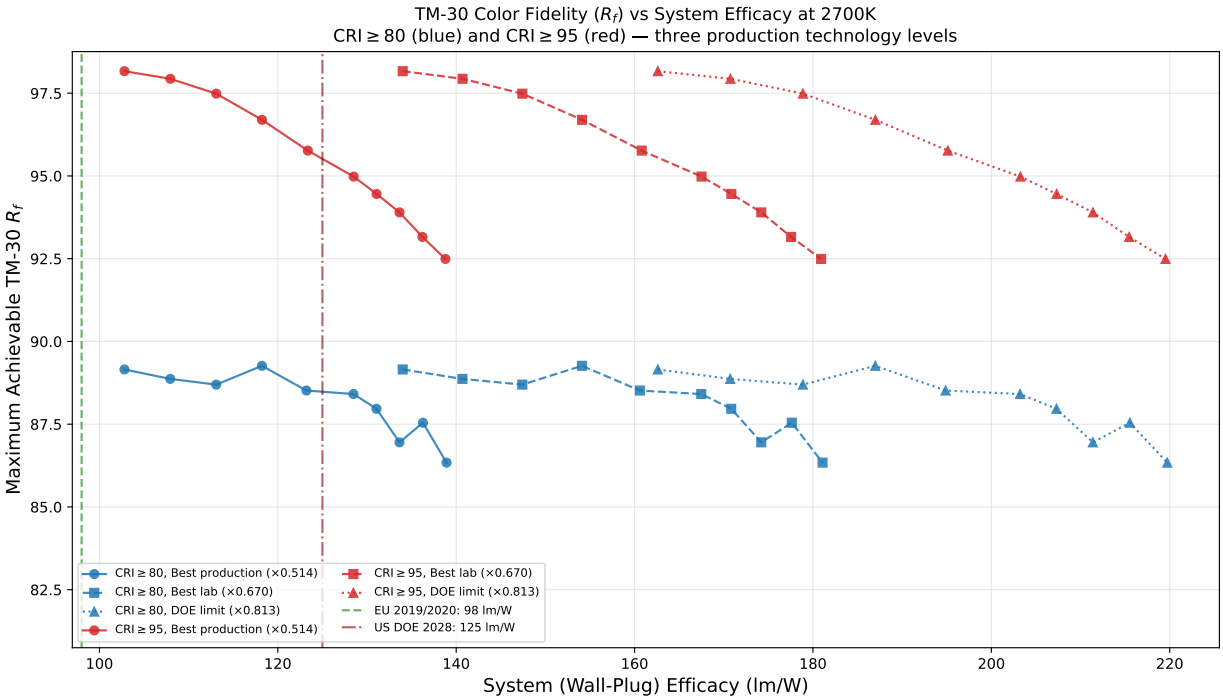

Figure 12: Combined view: maximum R_f vs. system efficacy for both CRI ≥ 80 (blue) and CRI ≥ 95 (red) at 2700K. CRI ≥ 95 curves sit $\sim$6–10 points higher in R_f across all technology levels.

Figure 12 overlays both CRI constraints. Key observations:

1. **TM-30 R_f is remarkably robust to efficacy pressure.** Even at CRI ≥ 80, the worst R_f observed is 86 — compared to R_9 values well below zero at the same operating point.

2. **The "CRI loophole" is invisible to R_f.** The three-peak spectra that achieve CRI=80 with $R_9 < 0$ still score $R_f \geq 86$. This is because R_f averages over 99 samples: even if a few hue bins are poorly rendered, the overall fidelity remains high.

3. **R_g exposes what R_f cannot.** The gamut jump at high efficacy (from $\sim$89 to $\sim$103) reveals spectral distortion that R_f alone would miss. This supports the TM-30 recommendation to report both R_f and R_g together, not R_f alone.

4. **CRI ≥ 95 narrows the gap.** With tight CRI, the optimizer cannot find loophole spectra, so R_f tracks CRI closely ($R_f \approx 0.97 \times$ CRI).

These results confirm that replacing CRI with TM-30 R_f in regulations would not, by itself, prevent the efficacy-driven spectral distortion documented in earlier sections.

12.5 Per-Hue-Bin Analysis: Why Averages Hide the Problem

Both R_f and R_g are aggregate metrics — R_f averages fidelity over 99 samples, R_g averages gamut area over 16 hue bins. As such, they are susceptible to the same averaging artifact that makes

CRI R_a a poor indicator: localized failures in specific hue regions are diluted by acceptable performance elsewhere.

TM-30 provides per-hue-bin fidelity values $\Delta R_{f,hj}$ across 16 bins (where $\Delta R_{f,hj} = 0$ is perfect fidelity). Examining these reveals the distortion pattern that R_f alone conceals:

- **Low efficacy** (200 Philips lm/W, CRI ≥ 80): All 16 bins are uniformly negative (-0.02 to -0.10), indicating mild desaturation across the board. The worst bin (h14, red–purple) is -0.103. The average $R_f = 89$ is a fair summary because the errors are uniform.
- **High efficacy** (270 Philips lm/W, CRI ≥ 80): The pattern *reverses sign* in some bins. Green–cyan bins (h7–h9) shift to $+0.05$ to $+0.09$ (oversaturated), while red bins (h1, h14–h16) shift to -0.03 to -0.09 (desaturated). Because positive and negative errors *cancel in the average*, R_f reports 86 — only 3 points worse than the uniform case. But specific object colors are rendered very differently from the reference illuminant: greens too vivid, reds washed out.

This sign-flip pattern is invisible to any averaged metric, whether CRI R_a (8 samples) or TM-30 R_f (99 samples). A regulation based on the *worst* per-hue-bin value — $\min_j R_{f,hj}$ — would be far harder to game, since the optimizer cannot compensate for one bad bin by improving another.

The vindication of R_9. This analysis retroactively explains why R_9 (CRI special test color TCS09, "strong red," Munsell 4.5 R 4/13) is such an effective canary for spectral distortion. The red region is precisely where three-peak spectra fail worst: the gap between the blue pump at 450 nm and the red peak at 610 nm leaves deep reds and red–purples with insufficient spectral support. TCS09's reflectance straddles this gap, making it maximally sensitive to the specific distortion mode that efficacy-driven optimization produces.

As a single additional number alongside R_a, R_9 may be more informative than R_f — it tests the *weakest point* of the spectrum rather than averaging over regions where the spectrum performs adequately. California's JA8 standard, which requires $R_9 \geq 50$ alongside $R_a \geq 90$, appears well-targeted: it directly constrains the failure mode that efficacy pressure creates, without requiring the full TM-30 apparatus.

The ideal regulatory approach would combine both strategies: TM-30's per-hue-bin breakdown (for comprehensive coverage) with a hard floor on the worst bin (to prevent the averaging loophole). Until TM-30 achieves widespread regulatory adoption, R_9 remains a pragmatic and effective stopgap.

13 Physiological Analysis: LMS Cone Excitation

The colorimetric metrics analyzed so far—CRI, TM-30 R_f/R_g, and individual R_i scores—are all based on the CIE tristimulus framework, which integrates spectral information through color matching functions. These functions are *linear* combinations of the true cone responses, preserving chromaticity but discarding information about the spectral distribution of cone stimulation. This section goes deeper: we compute the actual L, M, S cone excitations for several illuminants and analyze how the spectral structure of LED light differs from blackbody radiation at the level of individual photoreceptor channels.

13.1 Method

We use the Hunt-Pointer-Estevez (HPE) chromatic adaptation matrix to transform CIE 1931 $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ color matching functions into cone fundamentals:

$$\begin{pmatrix} L \\ M \\ S \end{pmatrix} = \mathbf{M}_{\text{HPE}} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}, \quad \mathbf{M}_{\text{HPE}} = \begin{pmatrix} 0.38971 & 0.68898 & -0.07868 \\ -0.22981 & 1.18340 & 0.04641 \\ 0 & 0 & 1 \end{pmatrix}$$

This physiologically-based transform (used in CIECAM97s) gives cone-like responses that are appropriate for chromatic adaptation modeling.

Von Kries adaptation is applied by normalizing each cone channel to the 2700 K blackbody white point: $L_a = L/L_{\text{ref}}$, and similarly for M_a and S_a. After adaptation, a perfect blackbody match gives $(L_a, M_a, S_a) = (1, 1, 1)$. Any deviation reveals a cone-level mismatch that the visual system may detect, even when tristimulus values (and hence CRI) are well-matched.

We quantify the deviation using the root-mean-square cone contrast (RMSCC):

$$\text{RMSCC} = \sqrt{\frac{1}{3}\left[\left(\frac{\Delta L}{L}\right)^2 + \left(\frac{\Delta M}{M}\right)^2 + \left(\frac{\Delta S}{S}\right)^2\right]}$$

13.2 Illuminants Compared

All illuminants are normalized to equal luminous flux ($Y = 1000$):

1. **2700 K Planckian radiator** — the reference.
2. **Waveform Lighting Centric Home A19** — a commercial CRI 95+, $R_9 = 91$ LED (digitized from published SPD, adjusted to match datasheet chromaticity $x = 0.4598$, $y = 0.4106$).
3. **CRI $\geq$ 80 max-efficacy** — our optimized spectrum with no R_9 constraint (highest LER).
4. **CRI $\geq$ 95 max-efficacy** — same with tighter CRI.
5. **CRI $\geq$ 80, Phil. 250** — Philips-corrected at 250 lm/W (achievable system efficacy ∼129 lm/W).
6. **CRI $\geq$ 95, Phil. 250** — same with tighter CRI.

13.3 Results: Von Kries Cone Contrast

Table 13: Von Kries adapted cone responses relative to 2700 K blackbody. $\Delta L/L$, $\Delta M/M$, $\Delta S/S$ are percentage deviations of each cone channel from the blackbody reference.

Source	$\Delta L/L$	$\Delta M/M$	$\Delta S/S$	RMSCC
2700 K Blackbody	0.00%	0.00%	0.00%	0.00%
Waveform A19	−0.50%	+0.35%	−3.71%	2.17%
CRI 80 (max eff.)	−0.32%	+0.22%	−54.89%	31.69%
CRI 95 (max eff.)	−0.42%	+0.29%	−47.04%	27.16%
CRI 80, Phil. 250	−0.48%	+0.33%	−50.35%	29.07%
CRI 95, Phil. 250	−0.68%	+0.47%	−25.14%	14.52%

Table 13 reveals a dramatic finding. The L and M cone deviations are tiny ($< 1\%$) for all illuminants—these long- and medium-wavelength cones see similar signals from any spectrum that matches the 2700 K chromaticity. But the **S cone tells a completely different story**: the efficacy-optimized spectra have **45–55% less S-cone stimulation** than the blackbody reference.

The commercial Waveform LED achieves only 3.7% S-cone deficit—a testament to its broad phosphor emission. To put this in context: the just-noticeable difference (JND) for the S-cone channel is typically 5–10%, much larger than for L or M cones, because S cones comprise only $\sim$6% of the cone mosaic and are absent from the central fovea [20]. The Waveform's 3.7% S-cone deficit is therefore **at or below the threshold of visibility** for direct illuminant comparison on a neutral surface—consistent with the perceptual near-equivalence that high-CRI commercial LEDs achieve.

The visibility of this deficit depends critically on the viewing condition. In *successive* comparison (one source at a time, adapted to each in turn), the S-cone JND after adaptation is 5–10%, and 3.7% would be near or below threshold. But in *simultaneous* comparison—both sources visible in the same visual field, as when an incandescent and an LED illuminate adjacent patches of the same wall—the situation changes dramatically. Chromatic adaptation still occurs, but the visual system adapts to a *single* compromise state (roughly the spatial average of the visual field), not independently to each illuminant. Neither patch is individually "normalized away" by adaptation; instead, both are rendered through the same adapted gain, and their difference survives as a spatial color signal. Simultaneous chromatic discrimination thresholds along the S-cone axis are approximately 0.5–1% [19], an order of magnitude lower than the successive threshold, because the task is now spatial comparison rather than comparison against color memory. The Waveform's 3.7% S-cone deficit is therefore **4–8× the simultaneous JND**, and should be **clearly visible** as a subtle color difference between the two patches of wall. The incandescent side would appear slightly warmer (more yellowish-pink), the LED side slightly more neutral, driven primarily by the S-cone channel.

Two additional mechanisms amplify the effect on real surfaces:

- *Deep red surface reflectance.* Wall paint is not spectrally flat; if the wall has any reflectance above 650 nm (as most warm-toned paints do), the blackbody's 5.2% deep-red L-cone contribution vs. the LED's 2.4% creates an additional L/M ratio difference that depends on the specific surface.

- *Spectral metamerism.* Surfaces with structured reflectance (wood, textiles, skin) may look quite different under the two illuminants even when the illuminants match on a neutral target.

The efficacy-optimized spectra produce S-cone deficits of 25–55%—**far above any visibility threshold**, whether simultaneous or successive. Even the "best" case (CRI 95, Philips 250 lm/W, 25% deficit) is dozens of JNDs. These spectra would look unmistakably different from incandescent, despite matching CRI and chromaticity: the S cone is being starved.

Figure 14 breaks down each cone's excitation by wavelength band, revealing how differently the illuminants distribute their spectral energy within each cone channel. The S-cone panel is most diagnostic: blackbody provides roughly equal contributions from the violet (420–460 nm) and blue (460–500 nm) bands, while LEDs concentrate S-cone stimulation in whichever band contains their pump emission.

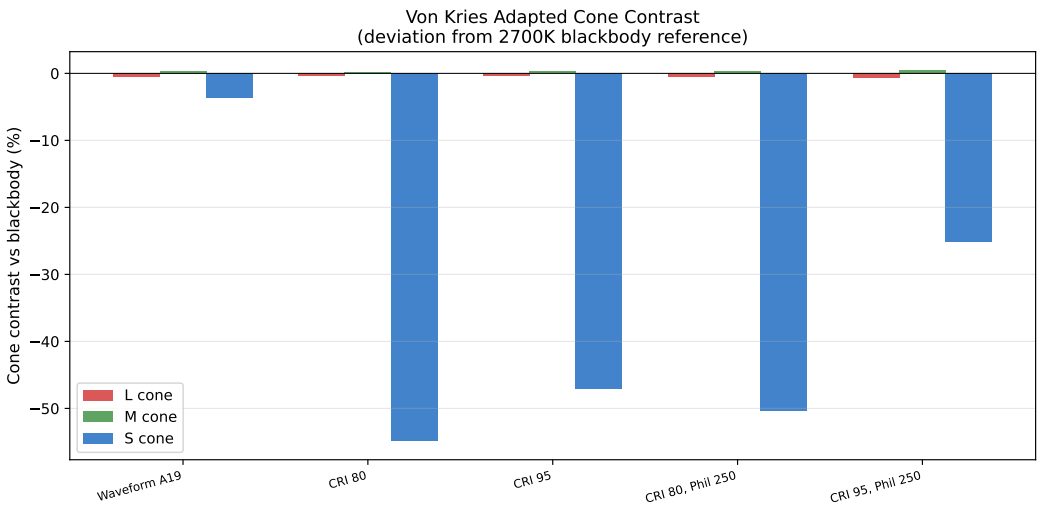

Figure 13: Von Kries adapted cone contrast versus 2700 K blackbody. The S-cone deficit (blue bars) completely dominates. The Waveform LED is nearly invisible; the optimized spectra show 25–55% S-cone deficiency.

13.4 Deep Red Contribution

A key spectral difference between incandescent and LED sources is the deep red region (> 650 nm). The 2700 K blackbody derives **5.2% of its L-cone excitation** from wavelengths above 650 nm, and a small but non-zero contribution from > 700 nm. By contrast:

- The Waveform LED gets 2.4% of L-cone from > 650 nm (half the BB).
- The efficacy-optimized spectra get $< 0.5\%$ — essentially zero deep red contribution.
- No LED spectrum provides *any* power above 700 nm, while the blackbody's Rayleigh-Jeans tail extends indefinitely.

This deep red deficit affects the rendering of surfaces with significant long-wavelength reflectance (wood, skin, brick, certain textiles). Even after von Kries adaptation normalizes the overall L-cone gain, surfaces that selectively reflect > 650 nm light will appear systematically desaturated under LED illumination.

13.5 Cone-Weighted Spectral Distribution

Figure 15 reveals *where* in the spectrum each cone gets its signal. For the L and M cones, the blackbody and LED distributions are broadly similar (both peak around 560–580 nm), but the blackbody's deep red tail (shaded region > 650 nm) provides a contribution that no LED replicates.

The most striking difference is in the S cone: the blackbody provides a broad, smooth signal from 400–500 nm, while the LED concentrates its S-cone stimulation in the narrow 450 nm pump band. Even though the chromaticity matches (the integrated S response can be tuned by adjusting pump intensity), the *spectral distribution* within the S-cone sensitivity range is fundamentally different. This difference becomes visible when viewing surfaces with structured blue reflectance—such as certain dyes, minerals, or artworks—where the narrow pump cannot substitute for the broadband blue of a thermal source.

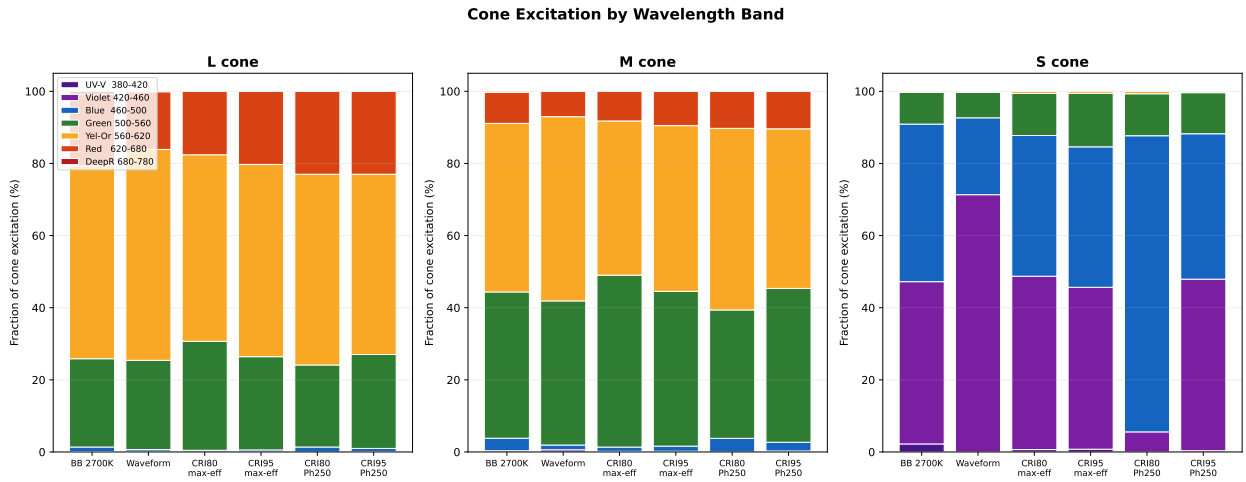

Figure 14: Cone excitation broken down by wavelength band (stacked bars). The L and M cone distributions are broadly similar across sources, but the S cone distribution (right panel) varies dramatically: blackbody provides balanced violet+blue, while LEDs concentrate S-cone stimulation in the pump band.

13.6 Five-Channel Receptor Analysis

The three-channel (L, M, S cone) analysis above captures chromatic differences but ignores two additional photoreceptor systems that contribute to perception at typical indoor light levels:

1. **Rod photoreceptors** — sensitivity function $V'(\lambda)$ (CIE 1951), peaking at 507 nm. Rods are active at mesopic light levels (∼0.01–600 cd/m^2) and contribute to brightness perception, spatial vision, and—through rod-cone interaction—color appearance [21].

2. **Melanopsin-containing ipRGCs** (intrinsically photosensitive retinal ganglion cells) — sensitivity function $s_{\text{mel}}(\lambda)$ (CIE S 026:2018, Lucas et al. 2014), peaking at ∼490 nm *in vivo*. These cells drive circadian photoentrainment, pupillary light reflex, and—importantly—contribute to perceived brightness and spatial brightness constancy even at photopic levels [22, 23].

Both of these receptor systems peak in the 490–510 nm range—precisely where phosphor-converted white LEDs have their characteristic spectral valley between the blue pump (∼450 nm) and the phosphor emission onset (∼520 nm).

We extend the analysis by computing five-channel adapted responses, normalizing each channel independently to the 2700 K blackbody reference, and computing a five-channel RMSCC:

$$\text{RMSCC}_5 = \sqrt{\frac{1}{5}\left[\left(\frac{\Delta L}{L}\right)^2 + \left(\frac{\Delta M}{M}\right)^2 + \left(\frac{\Delta S}{S}\right)^2 + \left(\frac{\Delta R}{R}\right)^2 + \left(\frac{\Delta \text{Mel}}{\text{Mel}}\right)^2\right]}$$

Table 14 reveals the central finding of this section. The Waveform LED—which appeared nearly equivalent to blackbody in the three-channel analysis ($\text{RMSCC}_3 = 2.2\%$)—shows **22.8% rod deficit** and **25.9% melanopic deficit**, driving the five-channel RMSCC to **15.5%**, a factor of **seven** larger than the cone-only metric.

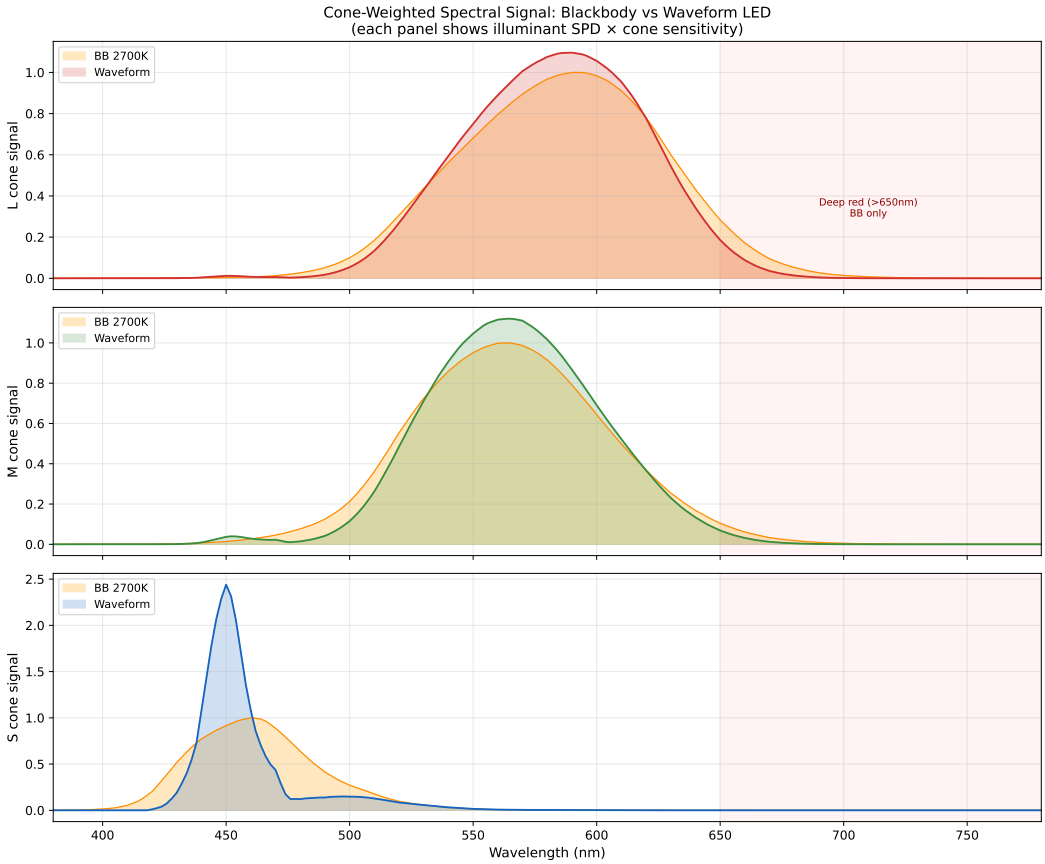

Figure 15: Cone-weighted spectral signal (illuminant SPD × cone sensitivity) for blackbody and Waveform LED. The deep red region (> 650 nm, shaded) contributes only to the L cone under blackbody illumination. The S-cone panel shows the Waveform's narrow pump versus the blackbody's broad blue distribution.

This result explains the perceptual observation that high-CRI LEDs, despite near-perfect chromaticity matching, still "feel different" from incandescent light—an effect that the colorimetric framework cannot account for.

13.6.1 Mesopic Vision in Residential Settings

The rod and melanopic deficits are not merely academic. The 2700 K warm-white LEDs that show these deficits are specifically designed for residential evening environments: bedroom lamps, living room ambient lighting, dining settings, and hospitality applications. These are precisely the conditions where *mesopic vision* dominates.

At typical residential evening illuminance levels (50–300 lux at the task plane), the corresponding retinal illuminance is well within the mesopic range (∼0.6–6 cd/m^2 for typical wall luminance at 20–30% reflectance). In this regime, rod photoreceptors are not saturated and actively contribute to:

- *Brightness perception* — mesopic brightness includes a rod contribution weighted by the CIE 1951 $V'(\lambda)$ function, which is exactly the channel showing a 23% deficit.

Table 14: Five-channel receptor contrast versus 2700 K blackbody. $\Delta R/R$ is the rod (scotopic) channel deficit; ΔMel/Mel is the melanopic channel deficit. RMSCC_3 uses only L, M, S; RMSCC_5 includes rod and melanopic channels.

Source	$\frac{\Delta L}{L}$	$\frac{\Delta M}{M}$	$\frac{\Delta S}{S}$	$\frac{\Delta R}{R}$	$\frac{\Delta \text{Mel}}{\text{Mel}}$	RMSCC_3	RMSCC_5
Waveform A19	−0.5%	+0.4%	−3.7%	−22.8%	−25.9%	2.2%	15.5%
CRI 80 (max eff.)	−0.3%	+0.2%	−54.9%	−43.1%	−50.3%	31.7%	38.5%
CRI 95 (max eff.)	−0.4%	+0.3%	−47.0%	−29.6%	−36.0%	27.2%	29.6%
CRI 80, Phil. 250	−0.5%	+0.3%	−50.4%	−21.7%	−22.6%	29.1%	26.5%
CRI 95, Phil. 250	−0.7%	+0.5%	−25.1%	−15.3%	−18.8%	14.5%	15.6%

- *Spatial vision and ambience* — rods dominate peripheral vision and contribute to the overall sense of "fill light" in a room. A 23% scotopic deficit means the LED-lit room provides less peripheral rod stimulation than the incandescent-lit room at the same photopic lux level.
- *Color appearance via rod-cone interaction* — at mesopic levels, rod signals feed into the chromatically-opponent pathways (the rod-bipolar/AII-amacrine pathway), influencing perceived hue and saturation [21].

The melanopsin-containing ipRGCs (26% deficit) further affect perceived brightness through their contribution to the pupillary light reflex and to the irradiance-coding pathway that modulates subjective brightness independently of the cone-based luminance channel [22].

The irony is striking: the spectral valley at 470–510 nm that enables efficient phosphor-converted white LEDs falls precisely on the sensitivity peaks of the two receptor systems most relevant to low-light comfort. The "cozy" warmth of incandescent light may owe as much to its rod and melanopic stimulation as to its chromaticity.

13.6.2 What a Lux Meter Misses

The scotopic-to-photopic (S/P) ratio and the melanopic daylight response factor (M-DLR, per CIE S 026:2018) quantify how much rod and melanopic stimulation a source provides per photopic lumen. Table 15 shows these ratios for all sources, along with the absolute scotopic and melanopic illuminance at a practical residential level of 200 photopic lux.

Table 15: Scotopic/Photopic ratio and Melanopic DLR. At 200 photopic lux (same meter reading for all sources), the absolute scotopic and melanopic illuminances differ dramatically.

Source	S/P	M-DLR	At 200 phot. lux		Deficit vs BB	
			Scot. lux	Mel. lux	ΔScot	ΔMel
2700 K Blackbody	0.882	0.391	176	78	—	—
Waveform A19	0.681	0.290	136	58	−23%	−26%
CRI 80 (max eff.)	0.502	0.195	100	39	−43%	−50%
CRI 95 (max eff.)	0.621	0.250	124	50	−30%	−36%
CRI 80, Phil. 250	0.690	0.303	138	61	−22%	−23%
CRI 95, Phil. 250	0.747	0.318	149	64	−15%	−19%

At 200 photopic lux—a typical residential evening level—the Waveform LED delivers **40 fewer scotopic lux** and **20 fewer melanopic lux** than a 2700 K blackbody at the same meter read-

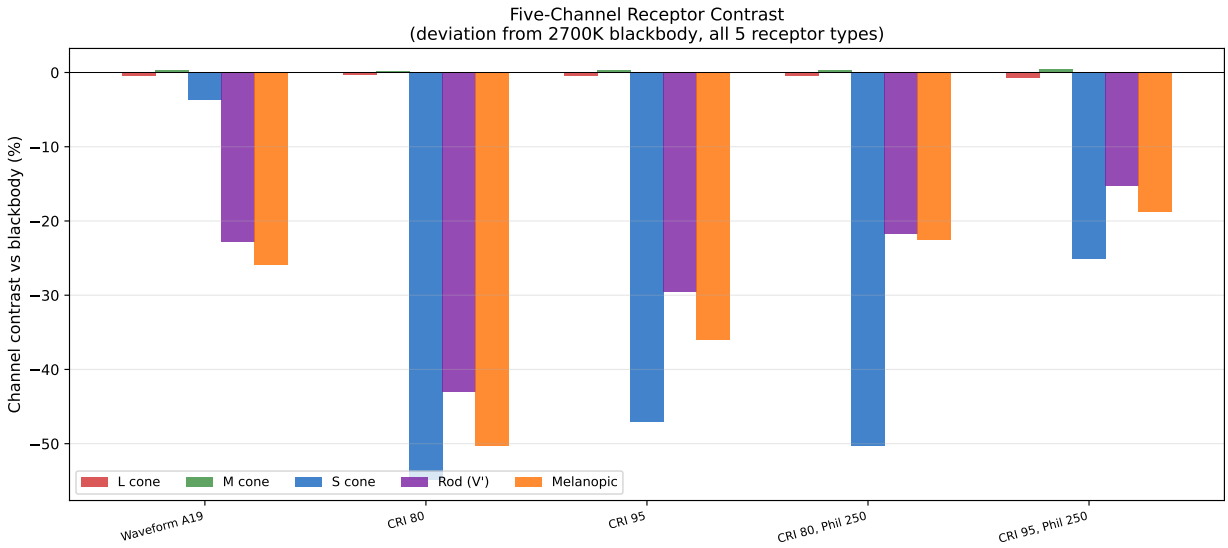

Figure 16: Five-channel receptor contrast versus 2700 K blackbody. The L and M cone deviations (red, green) are negligible for all sources. The S-cone deficit (blue) that dominated the three-channel analysis is dwarfed by the rod (purple) and melanopic (orange) deficits. Even the Waveform LED, with excellent colorimetric performance, shows ∼23–26% deficits in the non-cone channels.

ing. These are the missing lumens that no photopic instrument can detect but that the rod and melanopic photoreceptors respond to.

The CIE 191:2010 mesopic luminance model, applied to these S/P ratios, predicts only a 3% luminance deficit at 10 lux and effectively zero above 50 lux, because at typical residential wall luminances (∼5 cd/m^2 at 200 lux on a 30% reflectance surface) the CIE model considers vision to be purely photopic. However, the model was designed for road lighting and addresses *only* the luminance channel. Rod photoreceptors are physiologically active well above the CIE mesopic upper limit: rods do not fully saturate until ∼300 cd/m^2, two orders of magnitude above typical wall luminance. Their signals continue to contribute to spatial vision, peripheral ambience, and—through rod-cone interaction [21]—color appearance, through pathways that the CIE mesopic luminance model does not capture.

13.7 Implications

This analysis reveals a fundamental limitation of all colorimetric metrics (CRI, TM-30, individual R_i): they evaluate color rendering through tristimulus integration, which is blind to the spectral distribution of cone stimulation within each channel. Two spectra can have identical CRI and identical chromaticity, yet produce dramatically different cone excitation patterns.

The three-channel S-cone deficit is concerning but physiologically modest for high-CRI commercial LEDs (3.7% for the Waveform). The five-channel analysis reveals the true scale of the problem: even the best commercial LED shows a receptor-level mismatch seven times larger than cone-only metrics predict, driven entirely by the rod and melanopic channels.

Three aspects of this finding are particularly significant:

1. **Invisible to all existing metrics.** CRI, TM-30 R_f, and all chromaticity-based measures

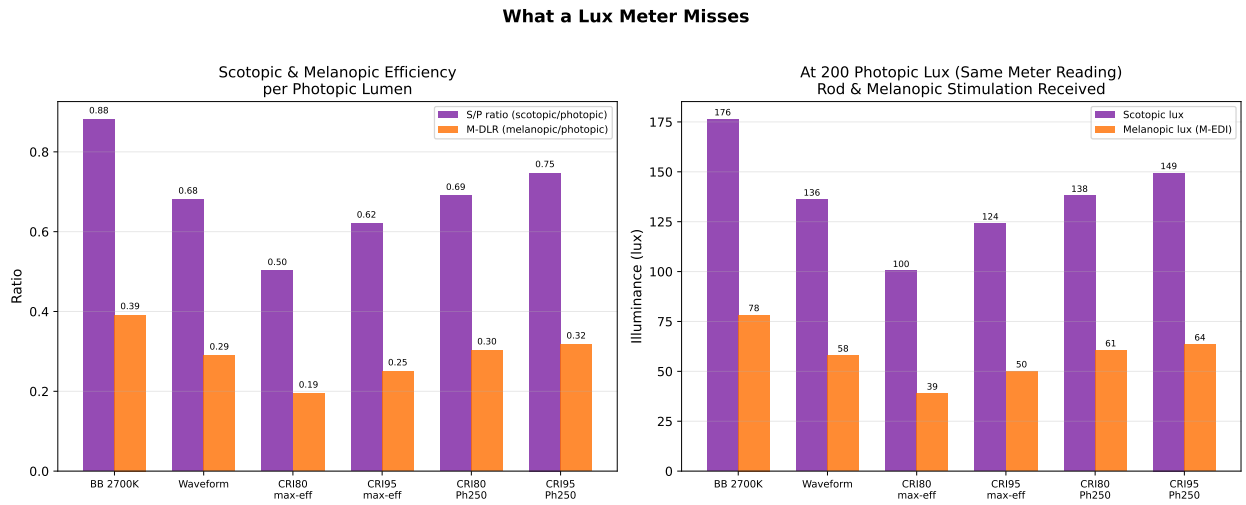

Figure 17: Left: S/P ratio and melanopic daylight response factor for each source. The blackbody provides the highest rod and melanopic stimulation per photopic lumen. Right: at 200 photopic lux (identical meter reading), the absolute scotopic and melanopic illuminance varies by a factor of $\sim$1.8$\times$ from best (blackbody) to worst (CRI 80 max-efficacy).

are defined on the CIE 1931 photopic observer ($V(\lambda) \approx 1.89L + M$). They contain no rod or melanopic information. A light source could have CRI 100, $R_f = 100$, perfect chromaticity, and still show 25%+ deficits in the non-cone channels.

2. **Worst where it matters most.** The rod and melanopic sensitivity peaks (490–507 nm) align with the LED phosphor valley. The 2700 K CCT products with the largest rod/melanopic deficits are marketed specifically for mesopic residential environments.

3. **No amount of spectral shaping within the phosphor band can fix it.** The pump-to-phosphor gap is a structural feature of phosphor-converted LEDs. Filling it requires either a cyan emitter ($\sim$490$-$510 nm) or accepting the loss of pump-conversion efficiency that filling this spectral region implies.

For the LED lighting industry, this suggests that a five-channel receptor analysis should complement traditional colorimetric metrics. A composite metric incorporating rod and melanopic channel contrast would capture the physiological reality that current metrics miss, and would specifically penalize the spectral valley that defines phosphor-converted white LEDs.

14 Future Work

Several directions for further work remain, building on the results reported above:

1. **TM-30 as optimizer constraint.** Section 12 maximizes R_f as the objective function while constraining CRI. The converse — constraining R_f while maximizing efficacy — would reveal what regulations written in TM-30 terms would actually require. A joint R_f/R_g constraint (e.g., $R_f \geq 85$, $R_g \leq 105$) appears from our results to close the CRI loophole. The R_{cs,h_j} per-hue-bin metrics could also be used as constraints to prevent selective desaturation of specific hue regions.

2. **Multiple CCTs.** Extend the sweeps of Section 9 to the full range of commercial color temperatures (2700–6500 K). Walter showed that the four-band positions are remarkably stable across ANSI lamp types; verifying this with our continuous parameterization and Philips cost model would be valuable.

3. **LED-realistic spectra.** Constrain the spectrum to physically realizable LED spectra: a narrow blue pump peak (∼450 nm, 20 nm FWHM) plus one or more phosphor emission bands (50–100 nm FWHM, constrained shapes). This would bridge the gap between our theoretical optima and practical LED design.

4. **Scotopic/mesopic optimization.** Section 13.6 reveals that 2700 K LEDs show 23% rod (scotopic) deficits versus blackbody, with the deficit concentrated in the 470–510 nm range. Adding a scotopic channel constraint to the optimizer—e.g., requiring $|\Delta V'/V'| \leq 5\%$—would force spectra to fill the pump-to-phosphor valley at the cost of some luminous efficacy. This is particularly relevant for residential evening lighting, where mesopic vision dominates and rod contributions affect perceived brightness and ambience.

5. **Melanopic-constrained optimization.** The 26% melanopic deficit (Section 13.6) is invisible to all current lighting metrics. Adding a melanopic equivalent daylight illuminance (M-EDI) constraint following CIE S 026:2018 would capture both the circadian dimension (melatonin suppression depends on melanopic stimulation) and the brightness/pupil dimension (ipRGCs modulate perceived brightness independently of cones). The trade-off between visual efficacy and melanopic adequacy could be mapped with the same sweep methodology used for CRI and R_f in this report.

6. **Optimizer robustness.** The CCT comparison (Table 2) reveals non-monotonicities where a tighter CRI constraint yields higher efficacy than a looser one, indicating that COBYLA finds different local minima at different constraint levels. Three improvements would help: (a) *neighbor seeding* — sweep CRI floors in order and use each solution as the starting point for the next, both upward and downward; (b) *multiple random restarts* per operating point, keeping the best result; (c) *monotonicity enforcement* — if a tighter constraint achieved higher efficacy, re-run the looser point seeded from the tighter solution, since relaxing a constraint can never worsen the true optimum.

15 Conclusion

The constrained optimization of light spectra reveals that regulatory pressure on luminous efficacy drives lamp spectra toward a mathematically determined "weird" final state: three narrow spectral peaks near 450, 540, and 610 nm with suppressed emission elsewhere. This state maximizes lumens per watt while technically satisfying color rendering requirements, but at the cost of poor rendering of saturated colors — especially reds.

The five-channel receptor analysis (Section 13.6) reveals a deeper problem: even well-designed commercial LEDs with CRI 95+ show 23–26% deficits in the rod and melanopic channels versus blackbody, invisible to all existing colorimetric metrics. These deficits are concentrated in the 470–510 nm spectral valley that is a structural feature of phosphor-converted white LEDs, and they fall precisely on the sensitivity peaks of the photoreceptors most relevant to the mesopic, residential environments where warm-white LEDs are deployed. This suggests that a complete characterization of light quality requires going beyond the three-cone colorimetric framework to include at least rod and melanopic receptor channels.

The `photopic` tool, built as a hybrid Modula-3/Scheme application, enables rapid exploration of this trade-off space using COBYLA constrained optimization with hierarchical spectral resolution.

Availability

The `photopic` tool, this report, the optimization data, and all plotting scripts are open source. The source code is available in two repositories:

- **Photopic tool and data:** `https://github.com/IntelLabs/async-toolkit`, branch `claude_edits`, path `m3utils/m3utils/photopic/`. The report and optimization results are in `photopic/report-runs/`.
- **CM3 compiler, MScheme, and runtime libraries:** `https://github.com/modula3/cm3`. MScheme lives in `m3-scheme/`; the COBYLA binding, spectral libraries, and CIE data are in the `m3utils` tree above.

Both repositories are licensed under open-source terms (Apache 2.0 for the async-toolkit; BSD for CM3).

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